

Box Robot Base

version 1.0, rev 18 29 December 2025

This is a classic box robot design intended to provide two forward facing attachments or a combination of forward and sidefacing attachments. We usually configure the left attachment motor for our worm drive attachment and the right attachment motor, rotated at 90 degrees, is usually for a geared lifting arm. We also use the right motor to drive a forward facing axle to turn objects on the FLL board (like the gears in Masterpiece Mission 12 rotating the chicken or Unearthed Mission 11/Angler Artifact).

We recommend using four large Spike Prime motors. If you only have two large motors use them for the attachments and use the medium Spike Prime motors for the wheels. Using the medium motors will require adjustments to the design of the front assembly section.

When assembling the robot care needs to be taken on how the sensor and motor cables need to be routed to the hub prior to joining the front and rear sections. Secure the cables using multiple wire connectors (49283) in the rear compartment containing the hub.

It is loosely based on:

<https://www.youtube.com/watch?v=PCMEvf4p2PI&t=2s>

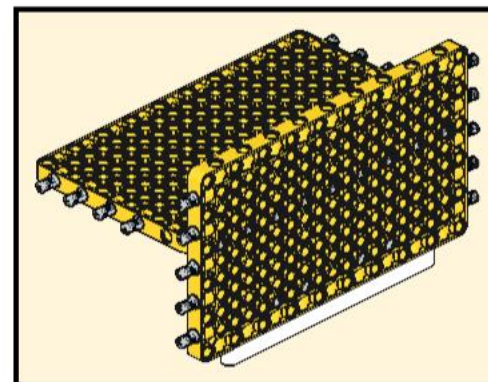
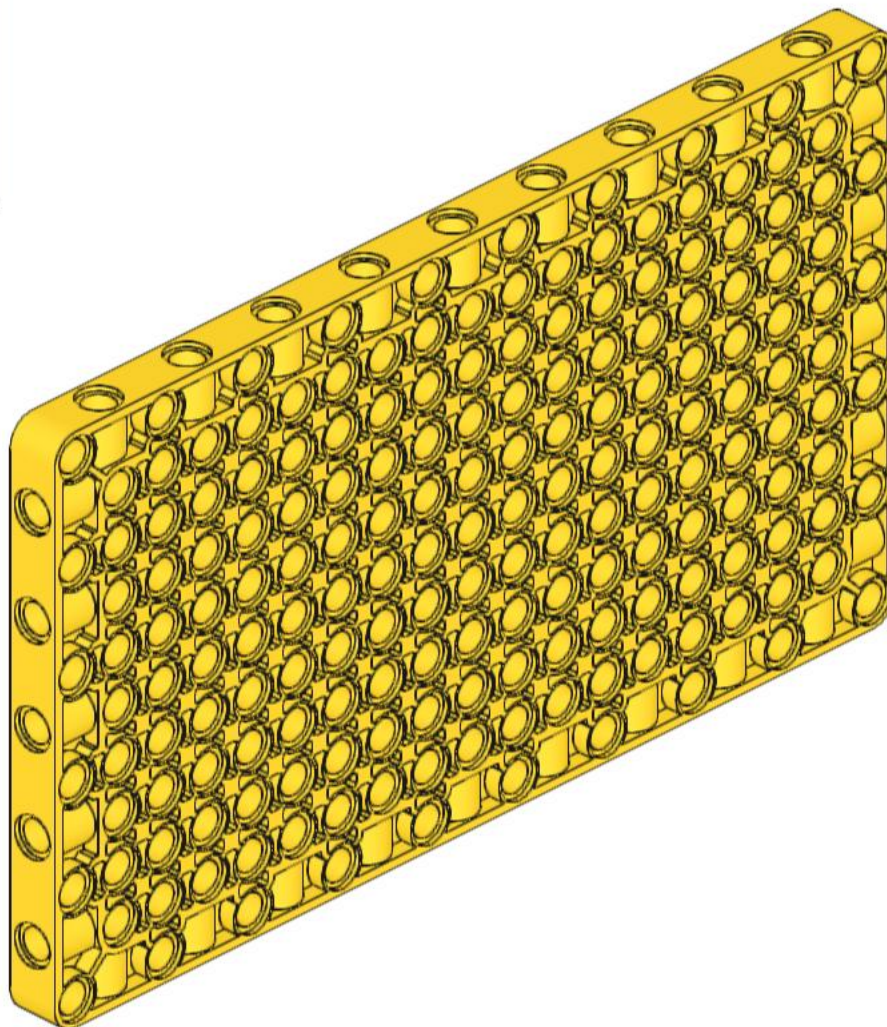
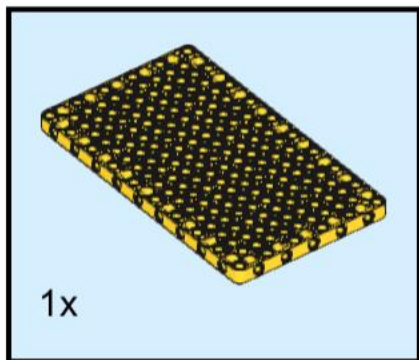
Front Assembly

Be careful on how attachment motors are placed on top of the yellow plate covering the driving motors.

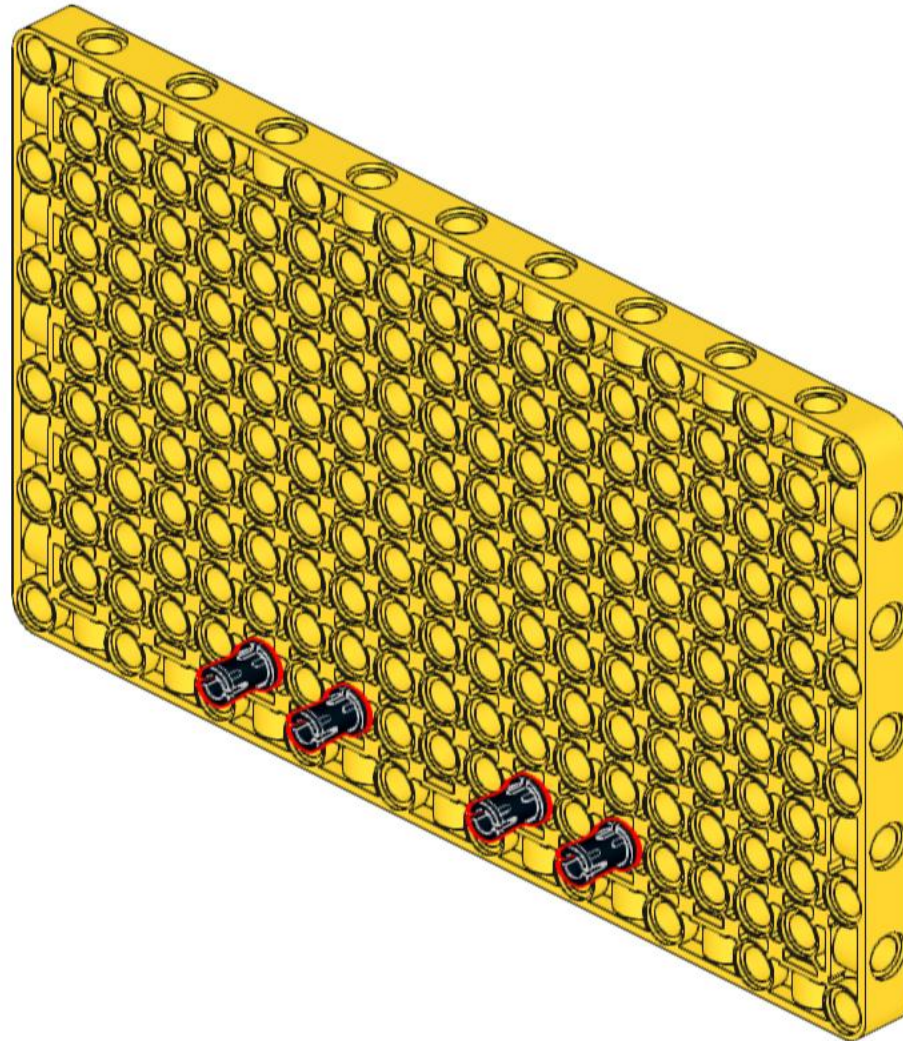
The primary mechanism the drive motors are secured is via attachment to this plate.

You need to optimize the number of pins (8) from the drive and attachment motors that are connected to the top and bottom of the plates.

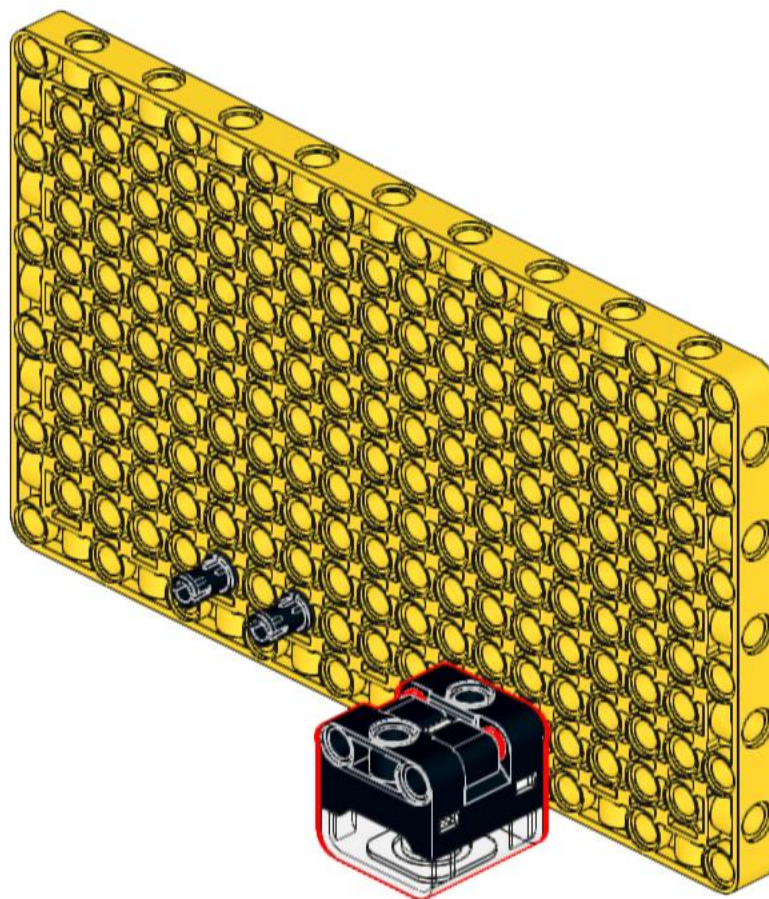
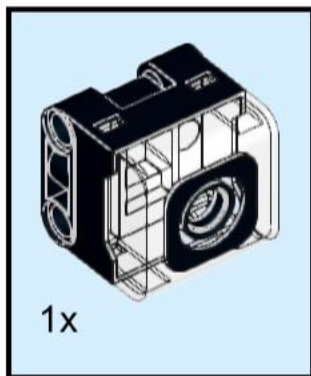
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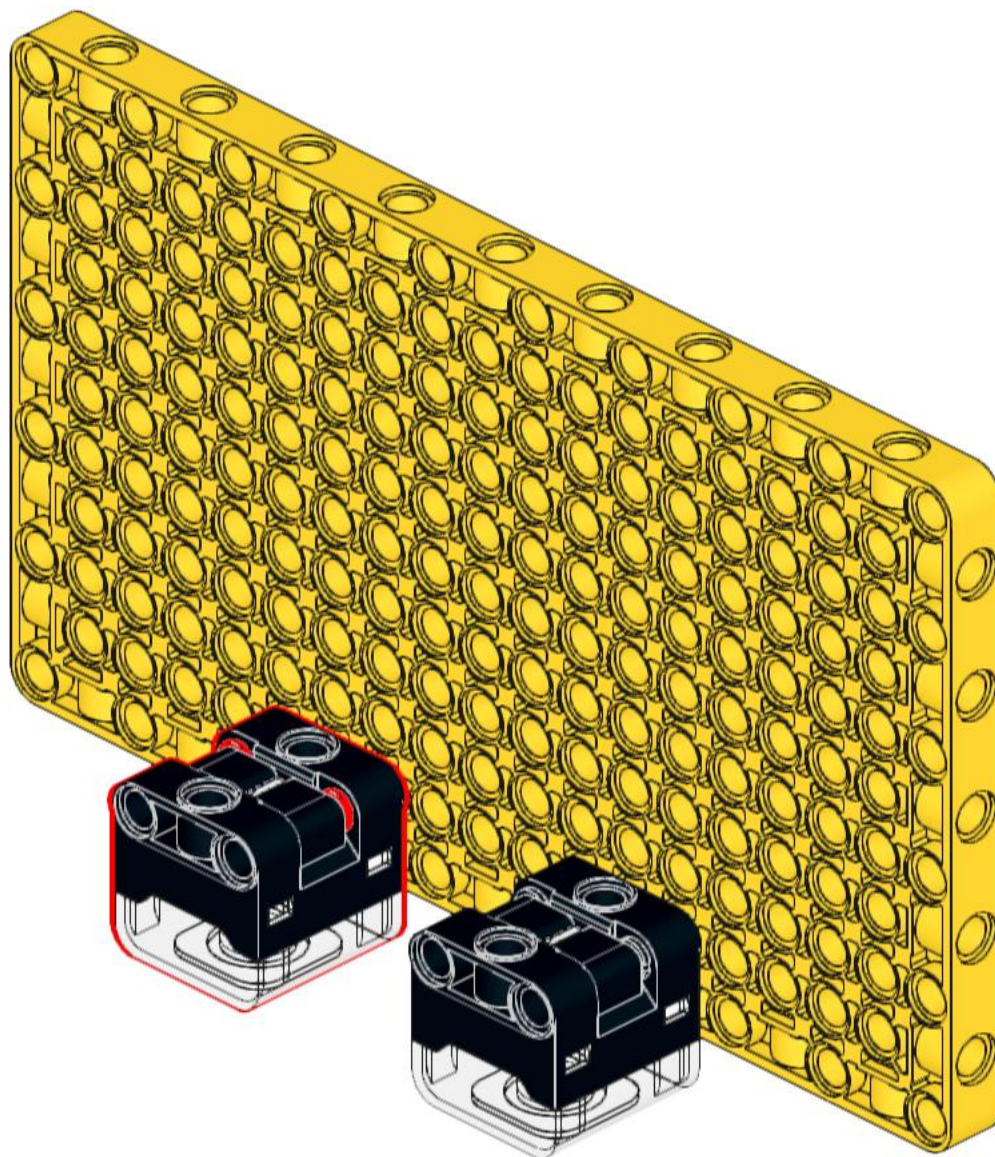
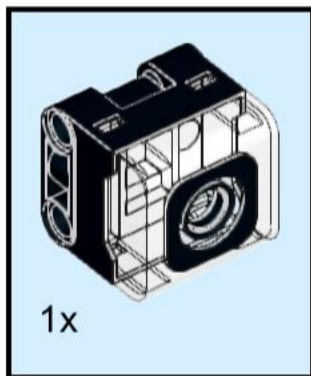
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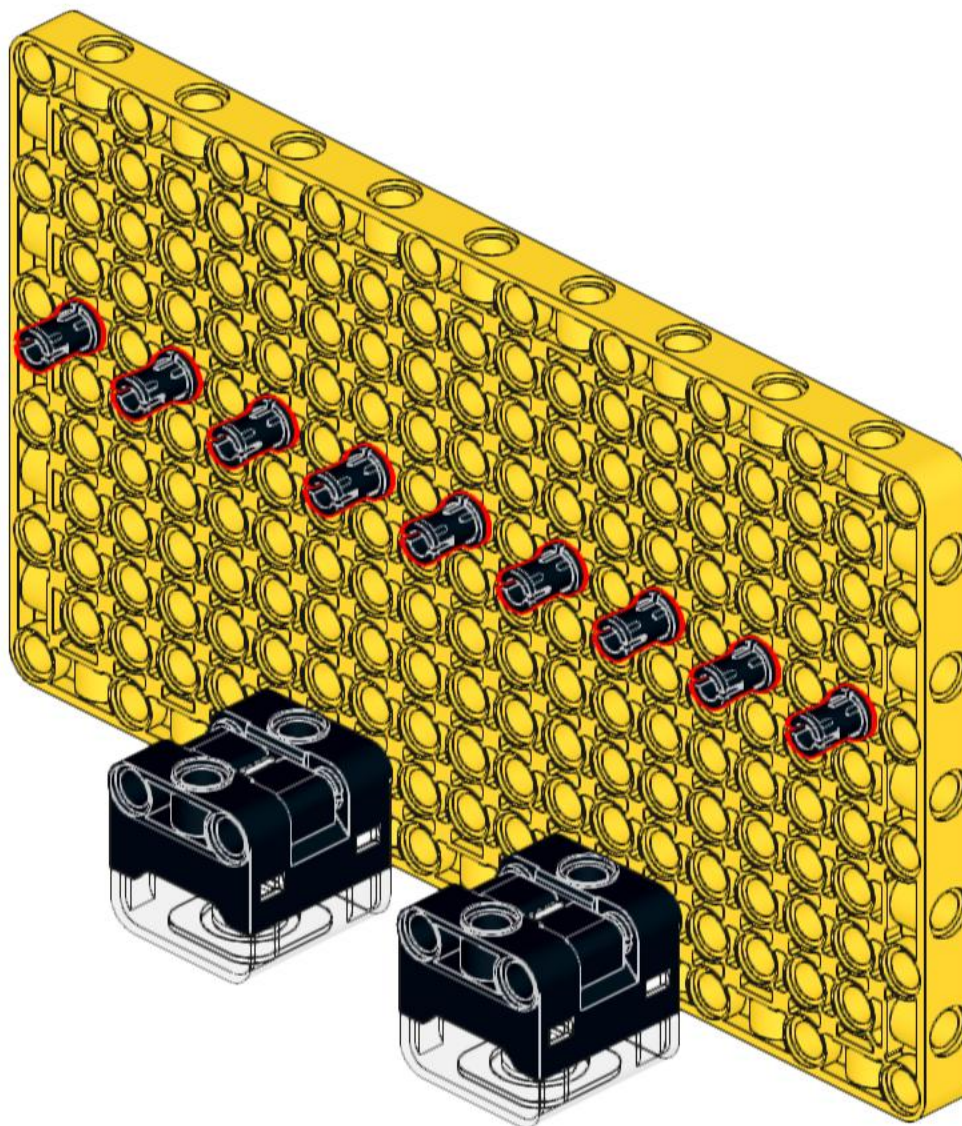
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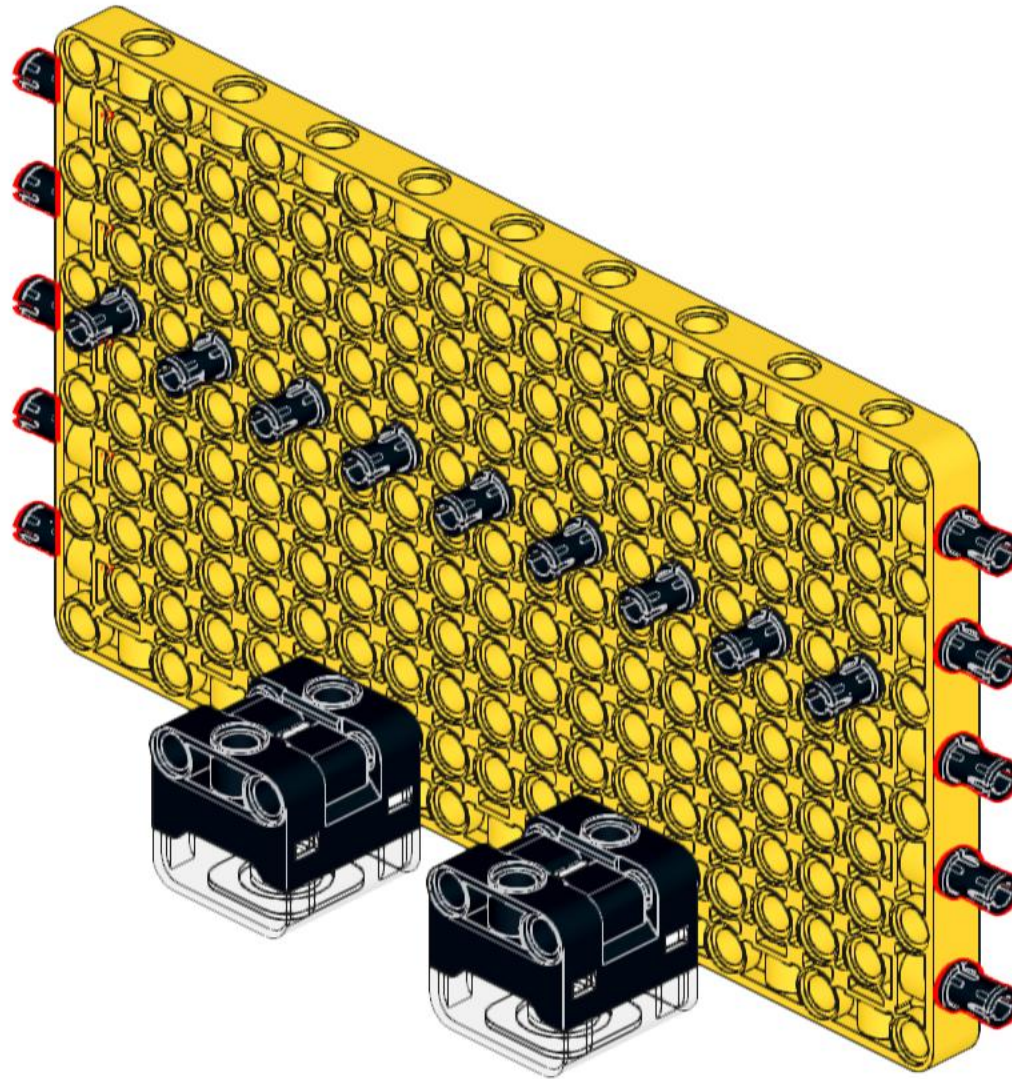
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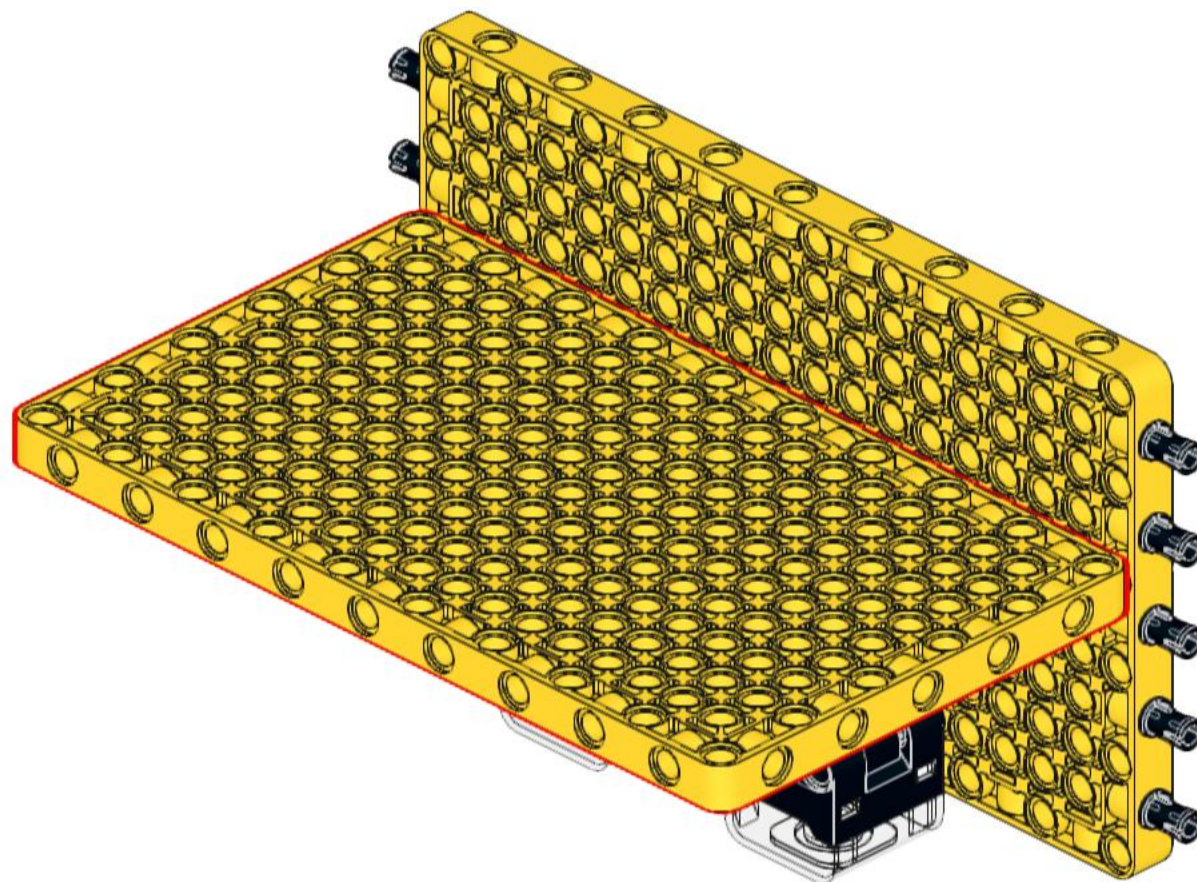
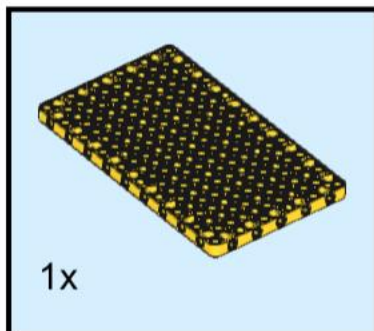
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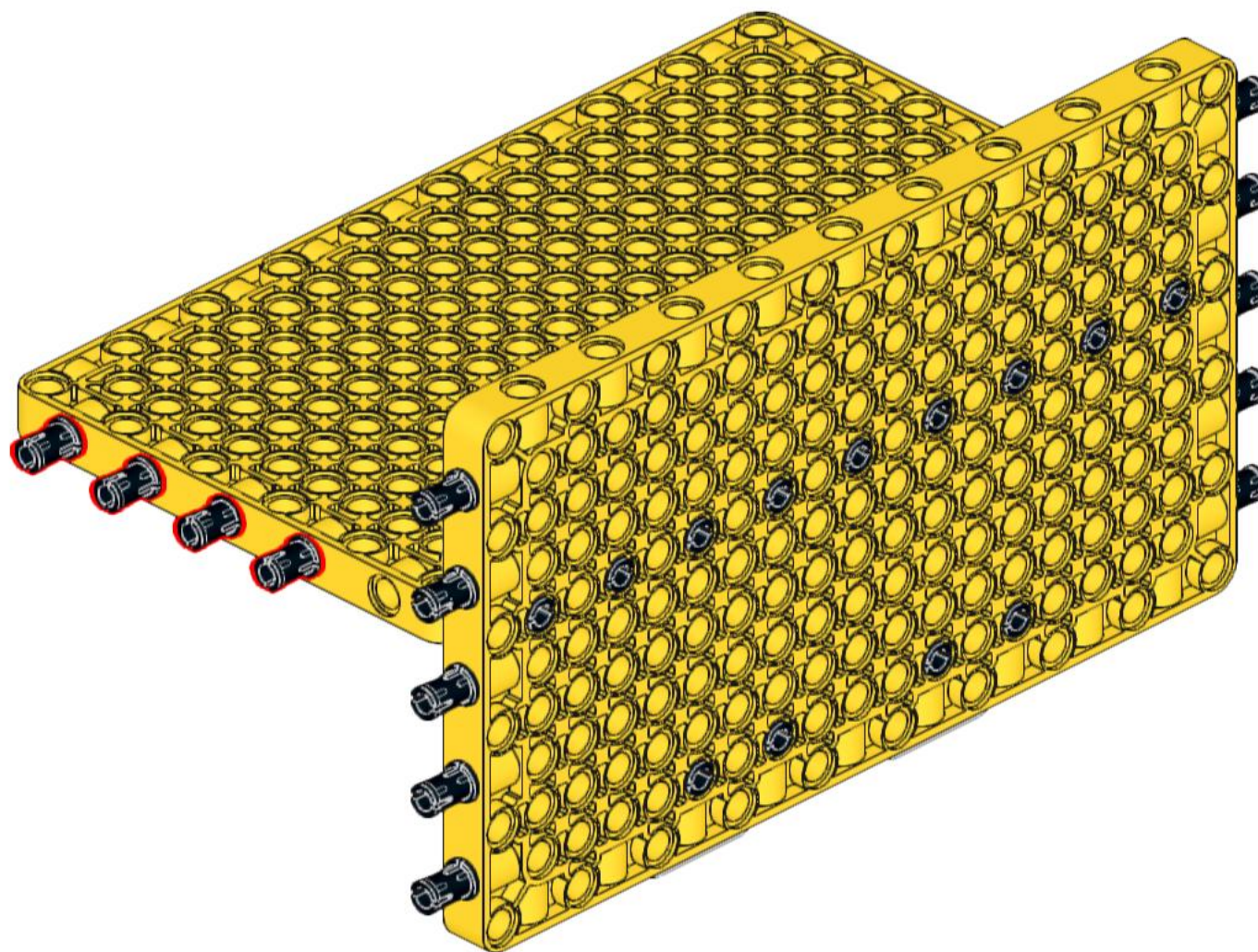
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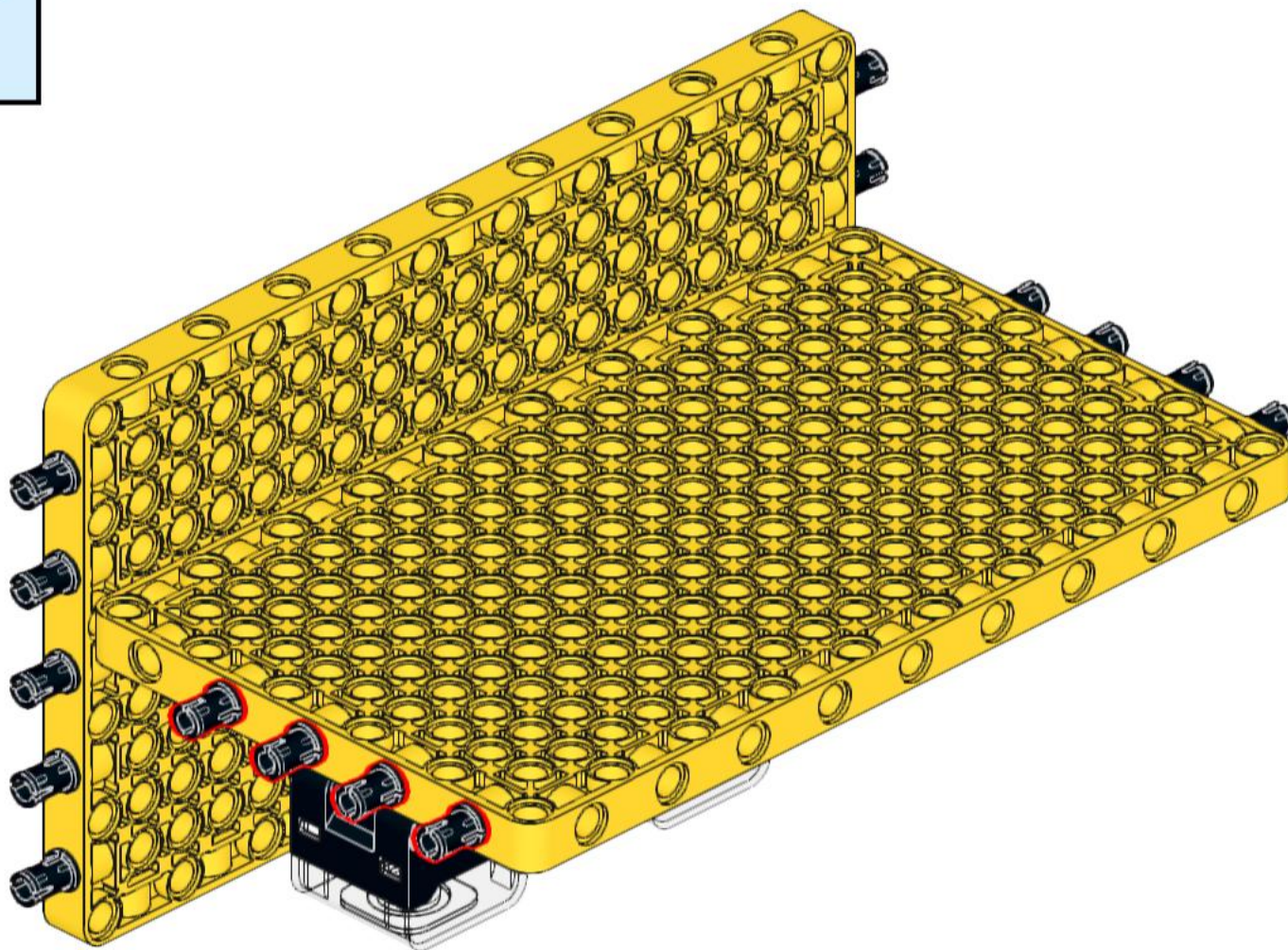
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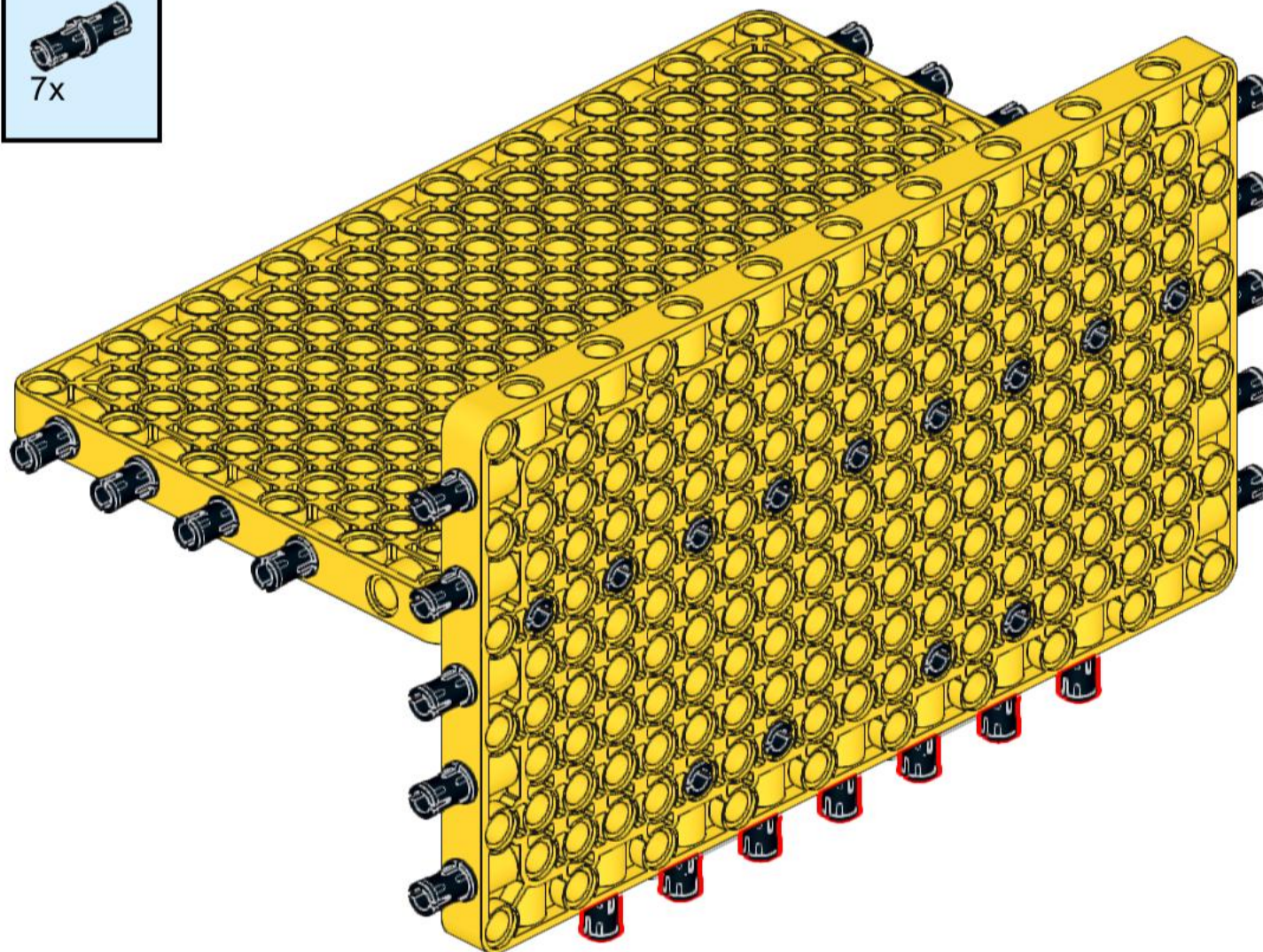
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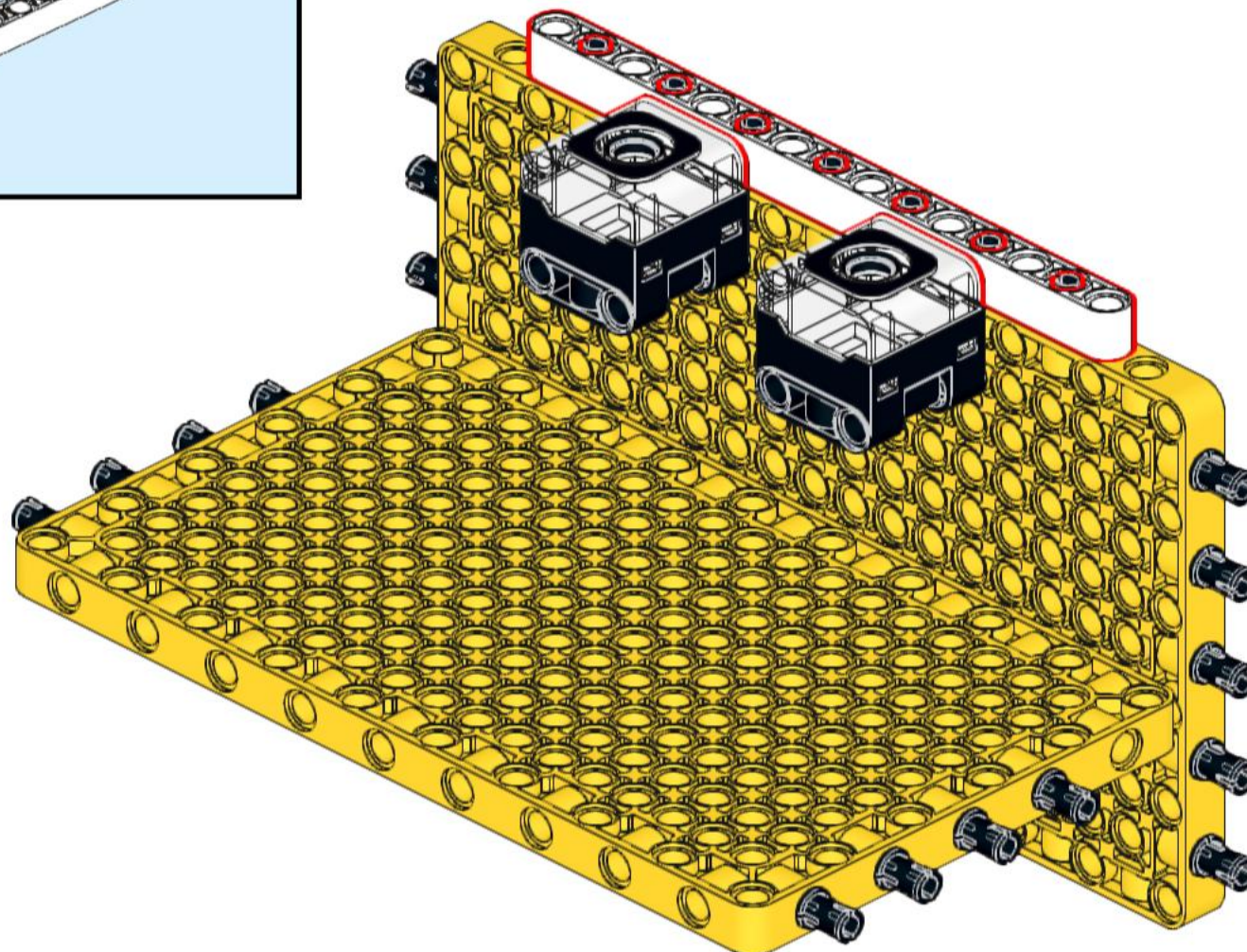
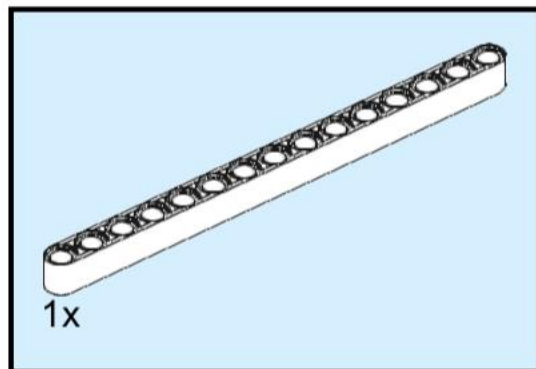
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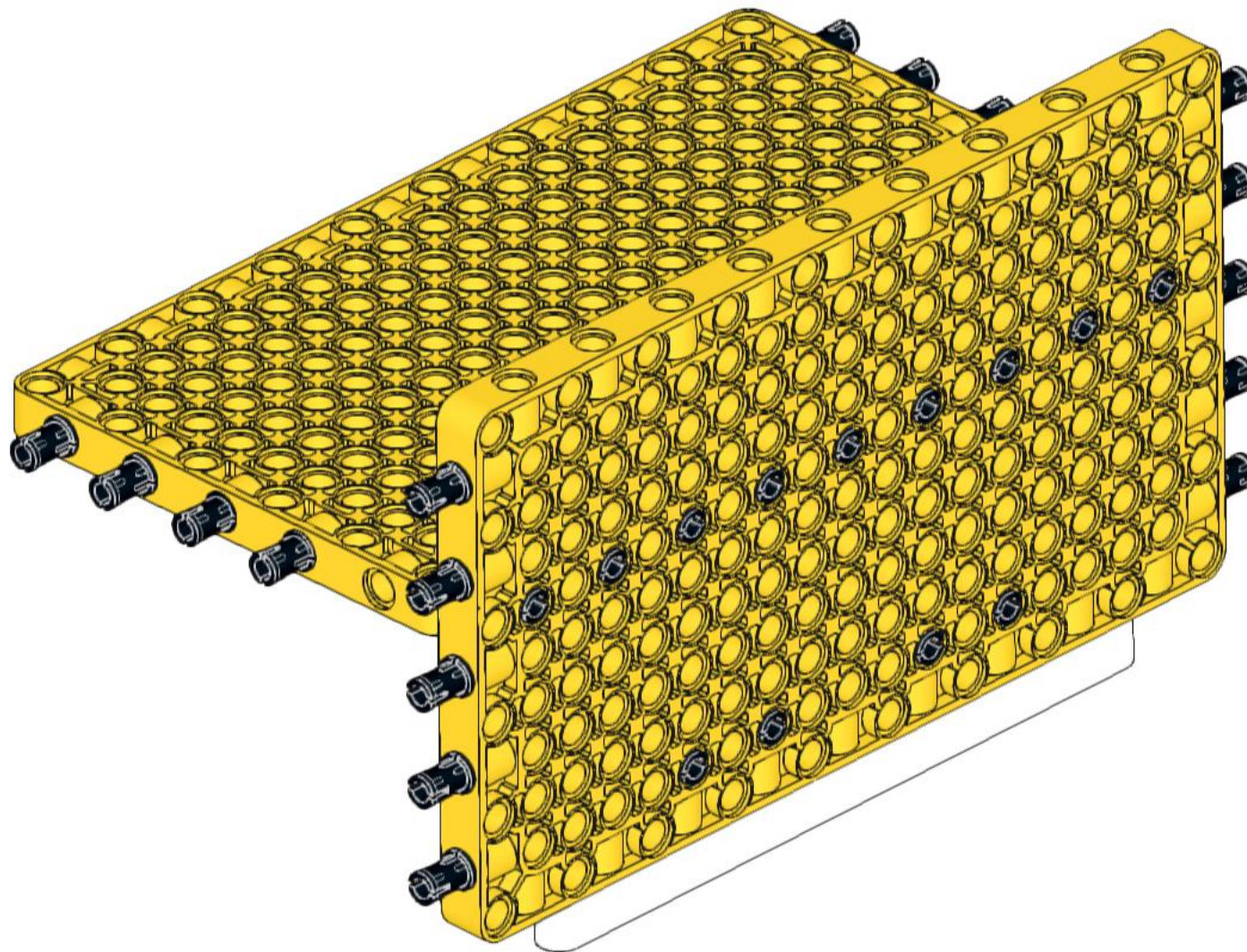
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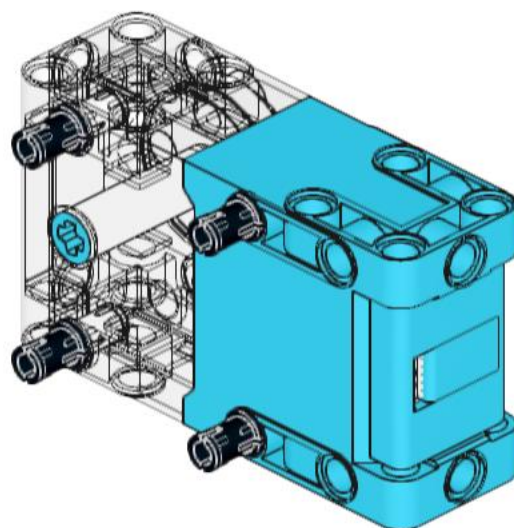
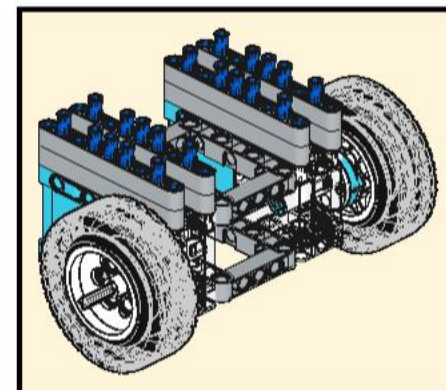
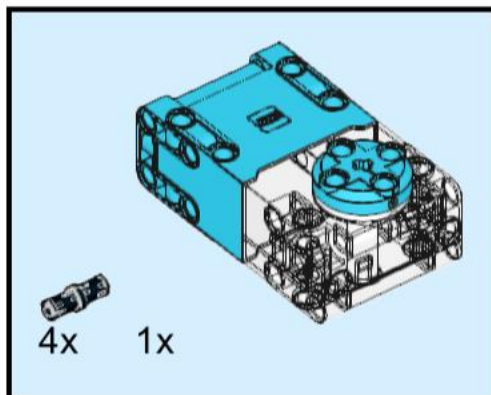
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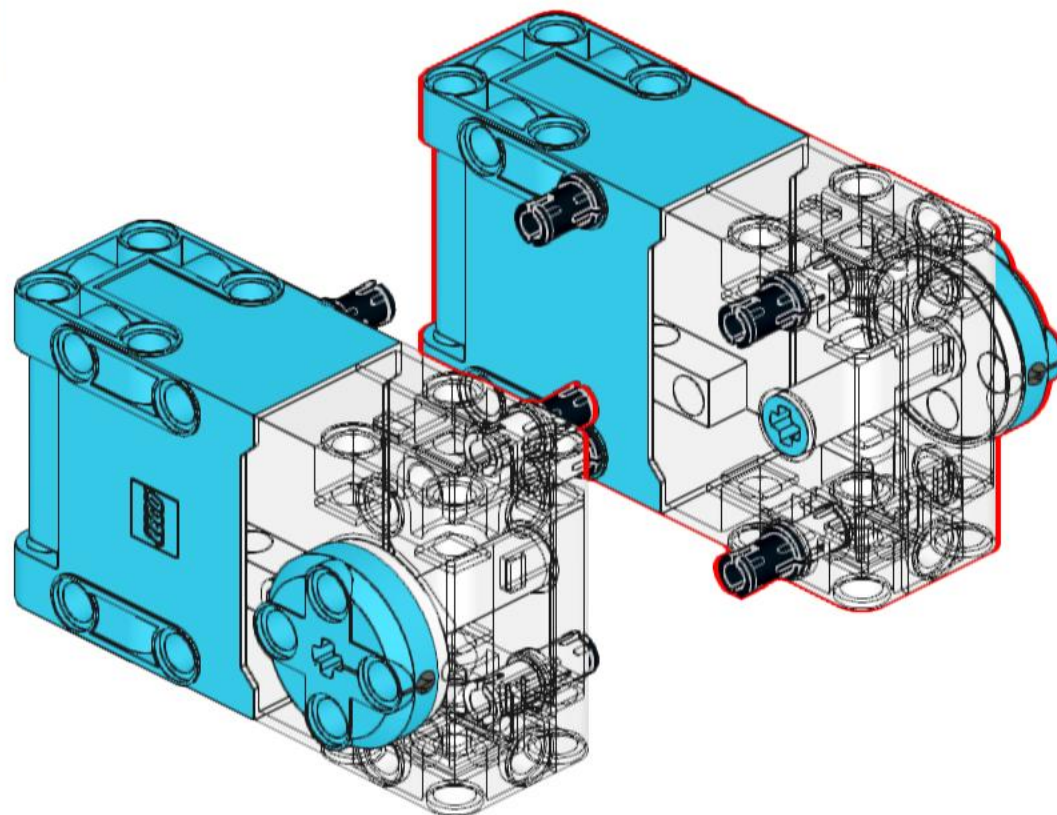
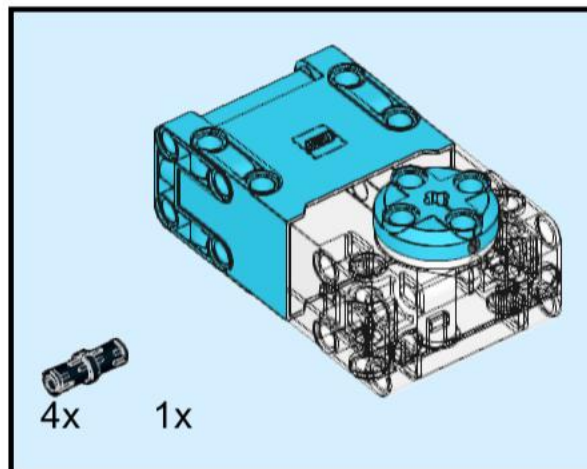
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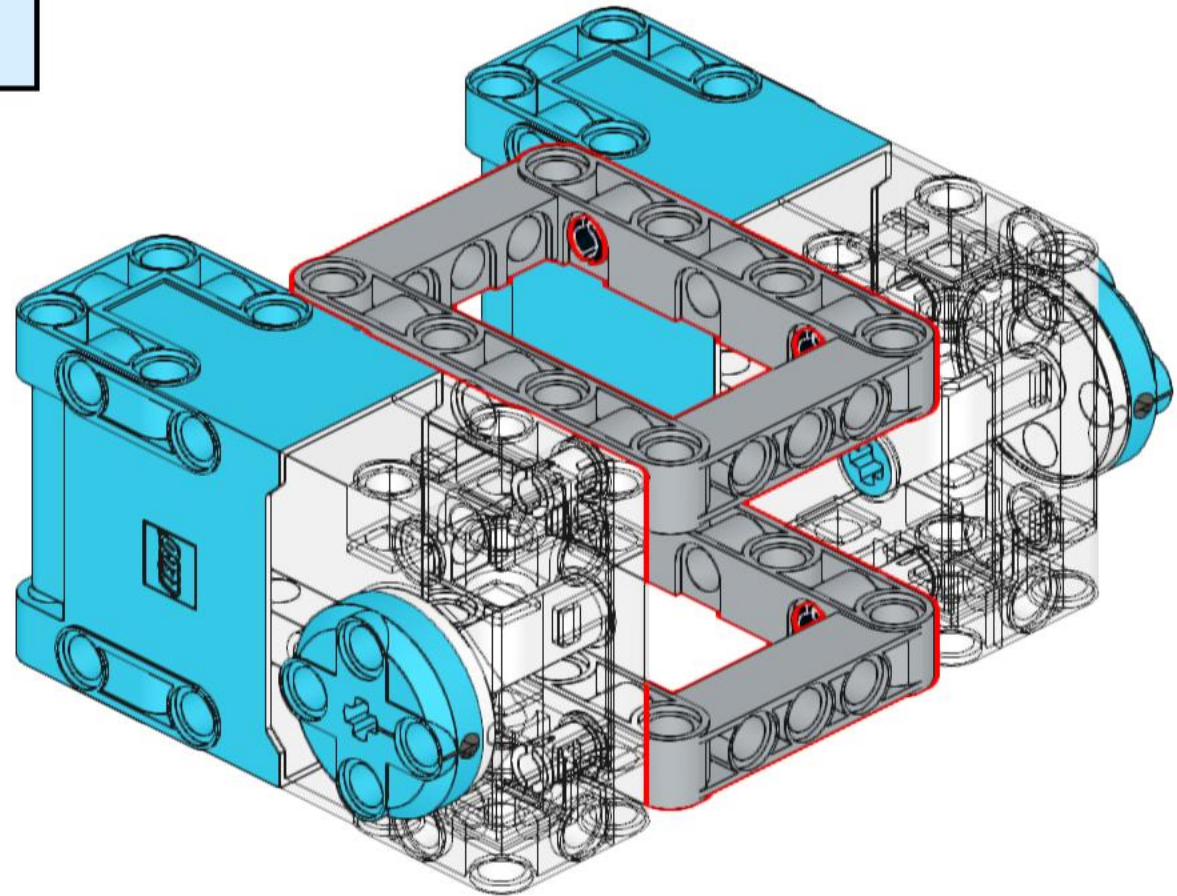
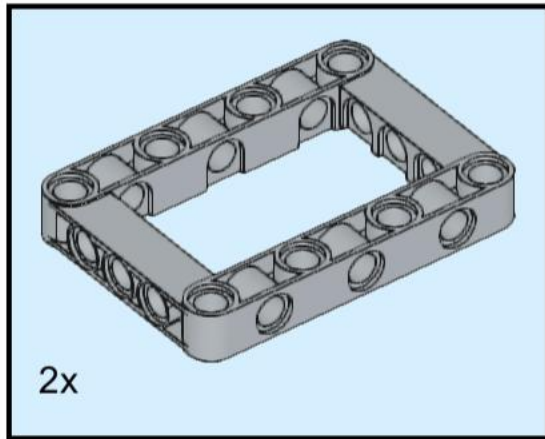
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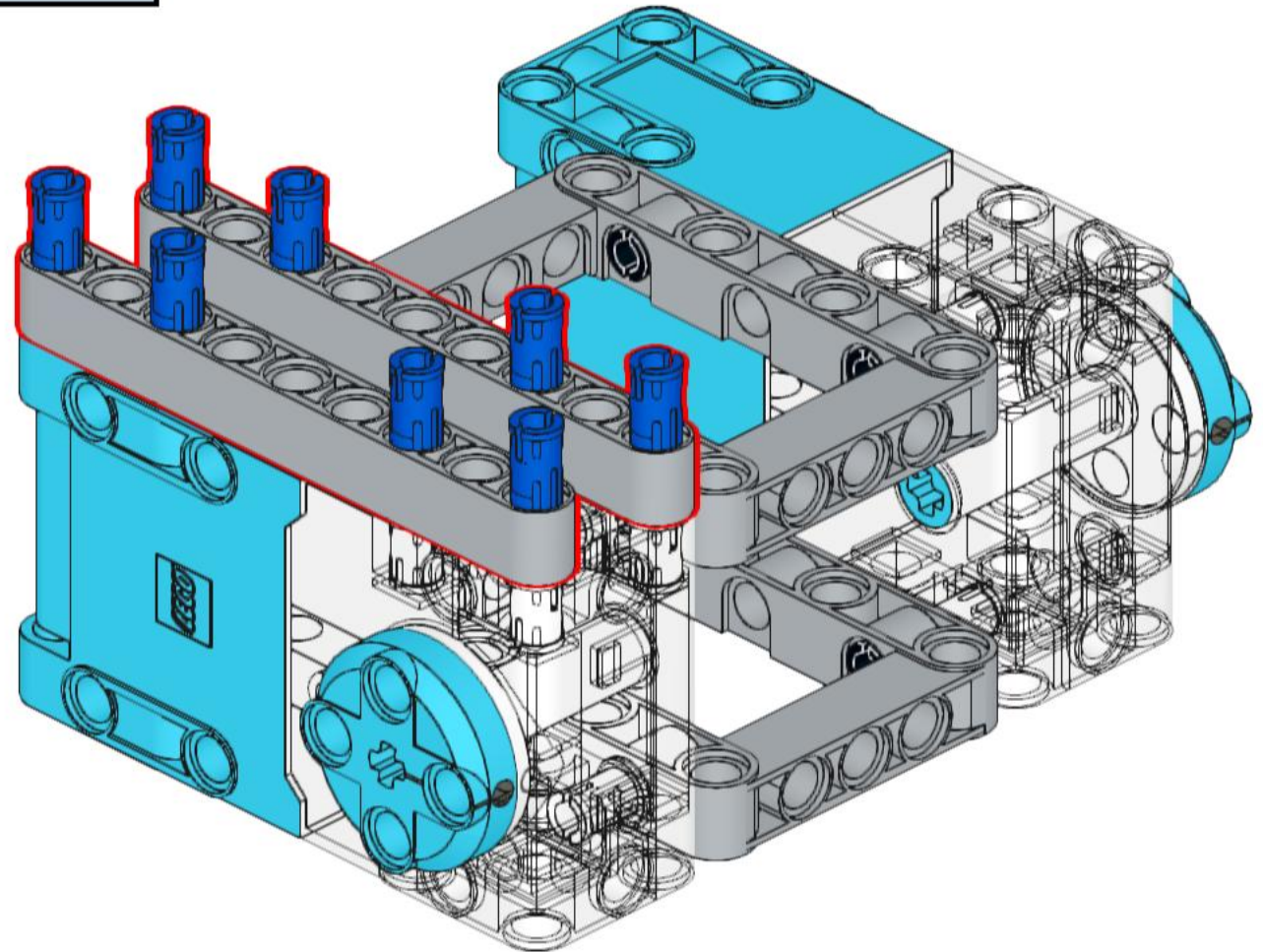
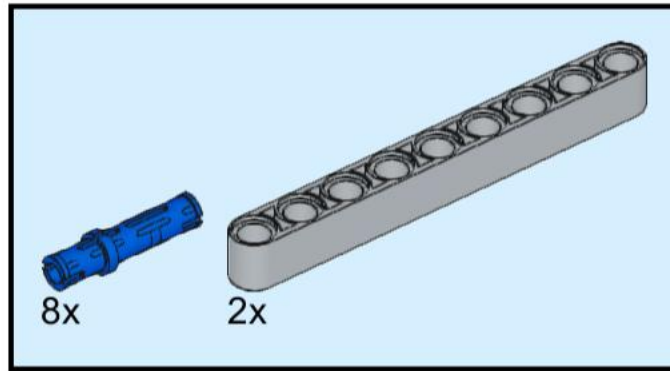
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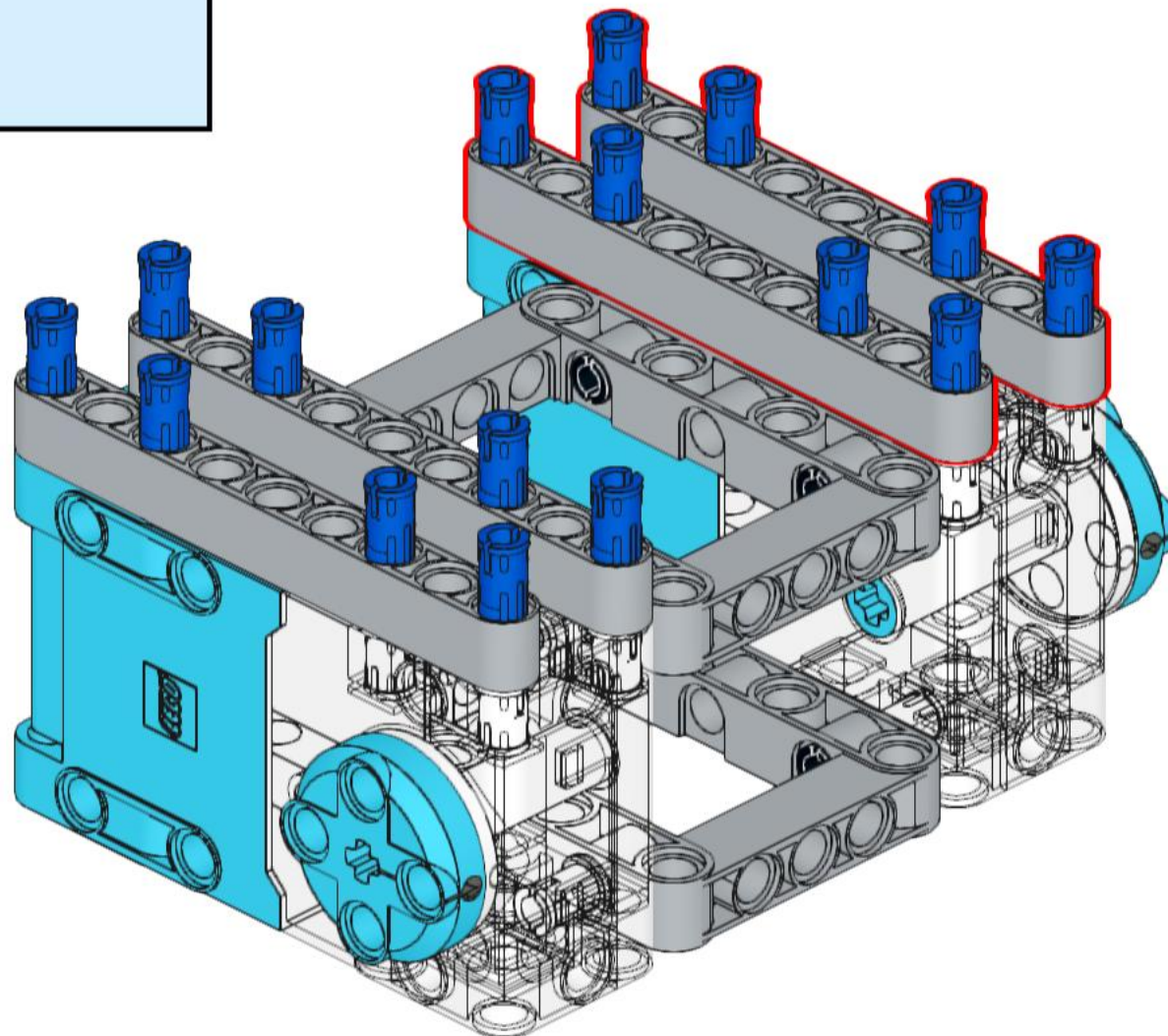
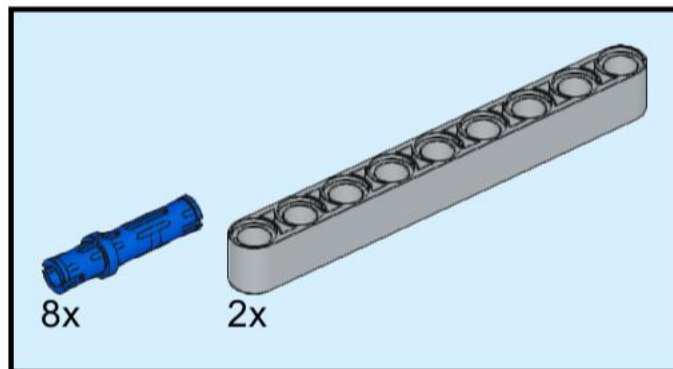
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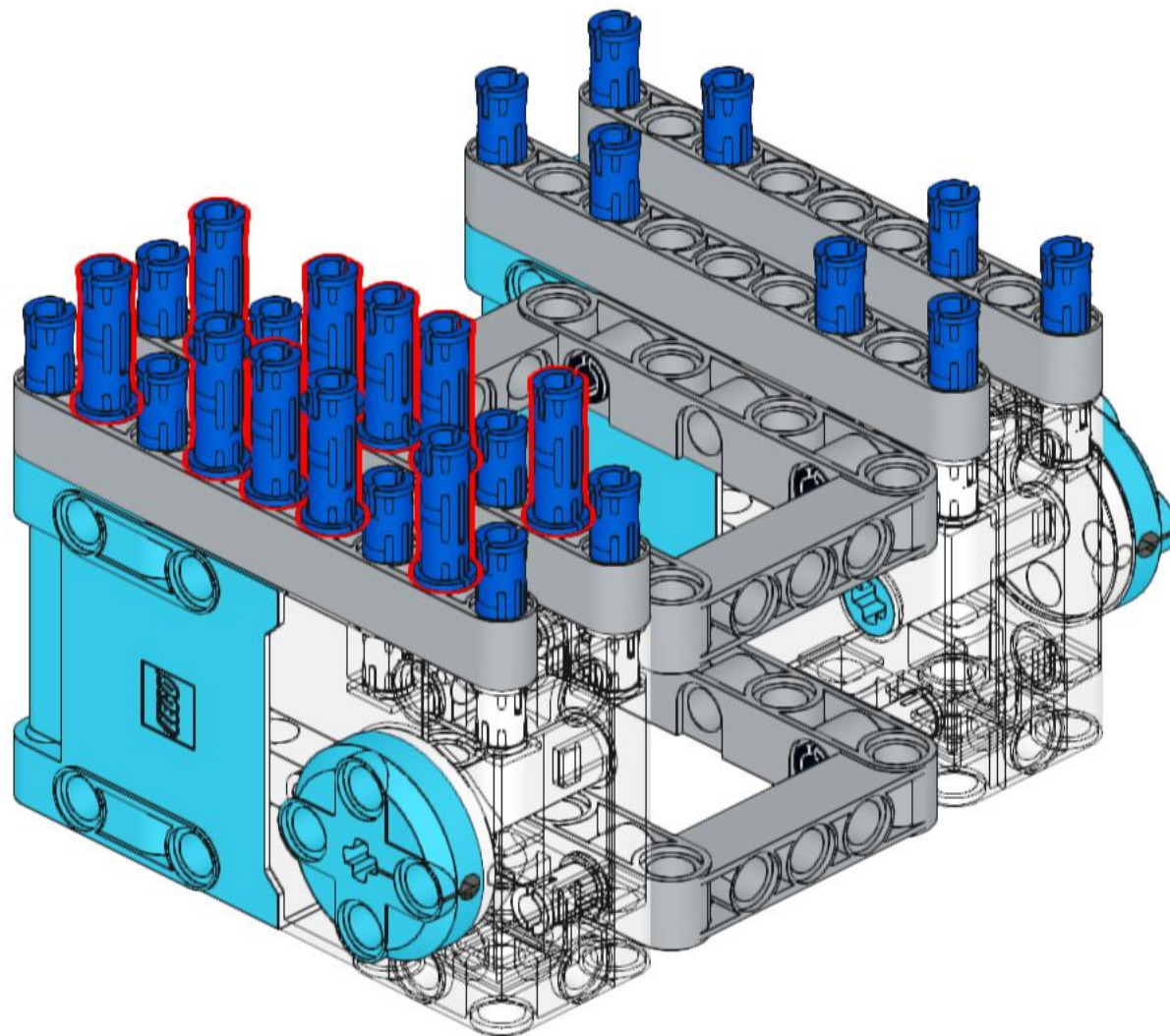
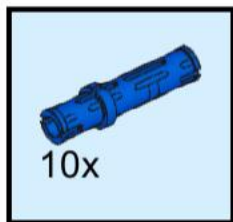
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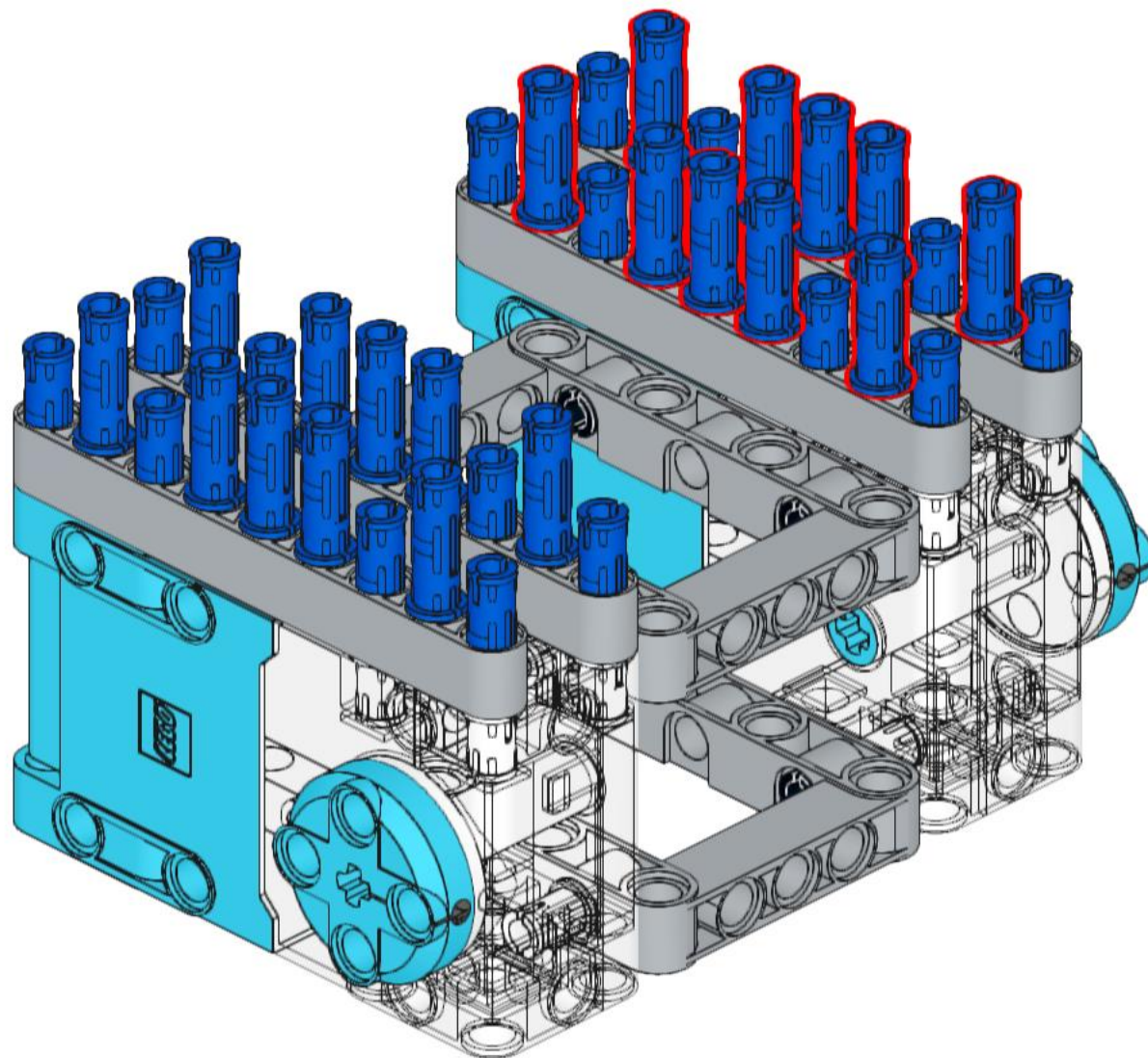
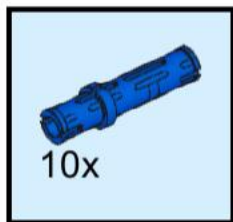
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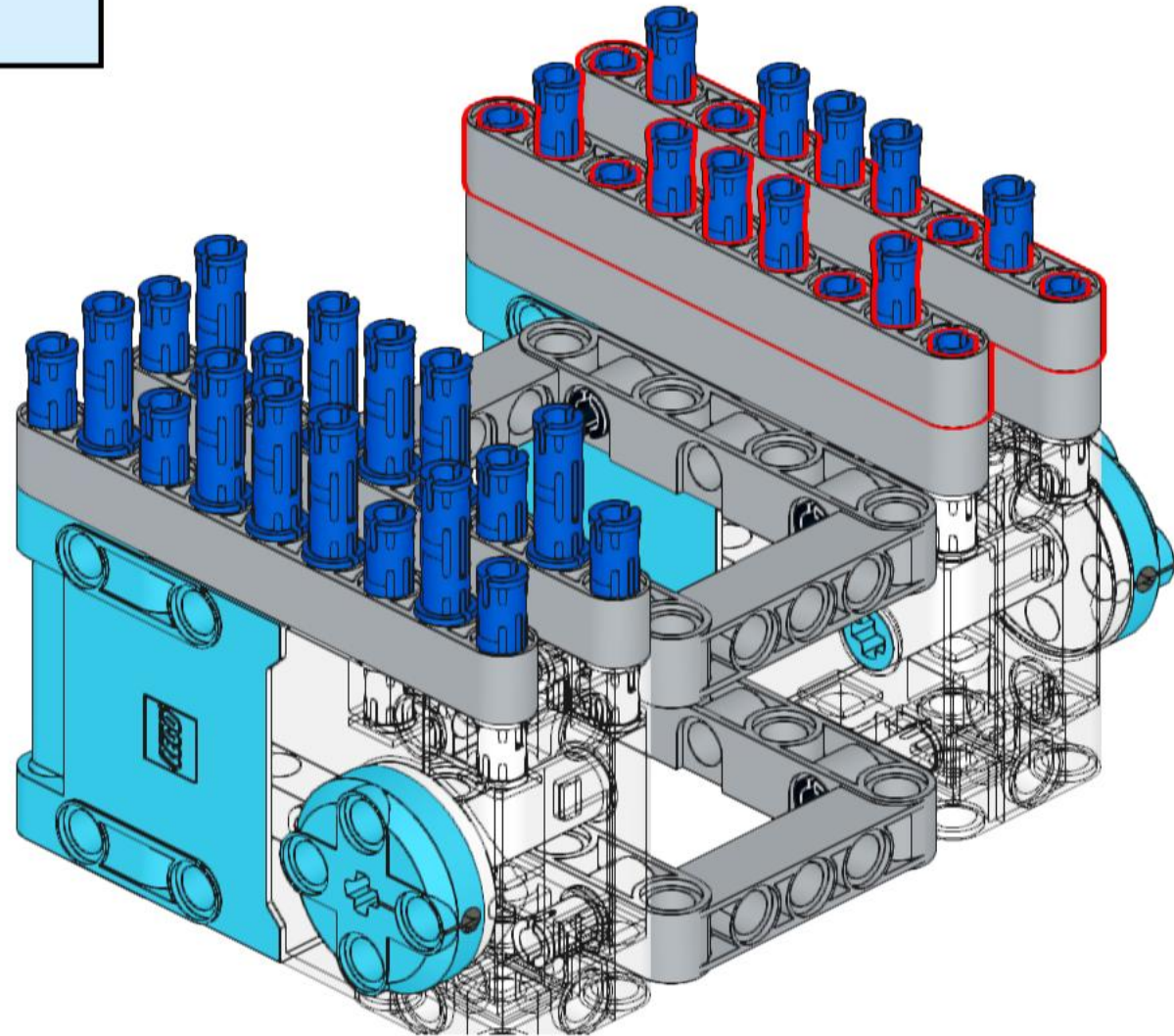
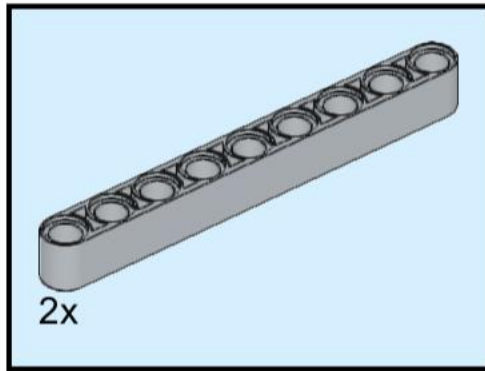
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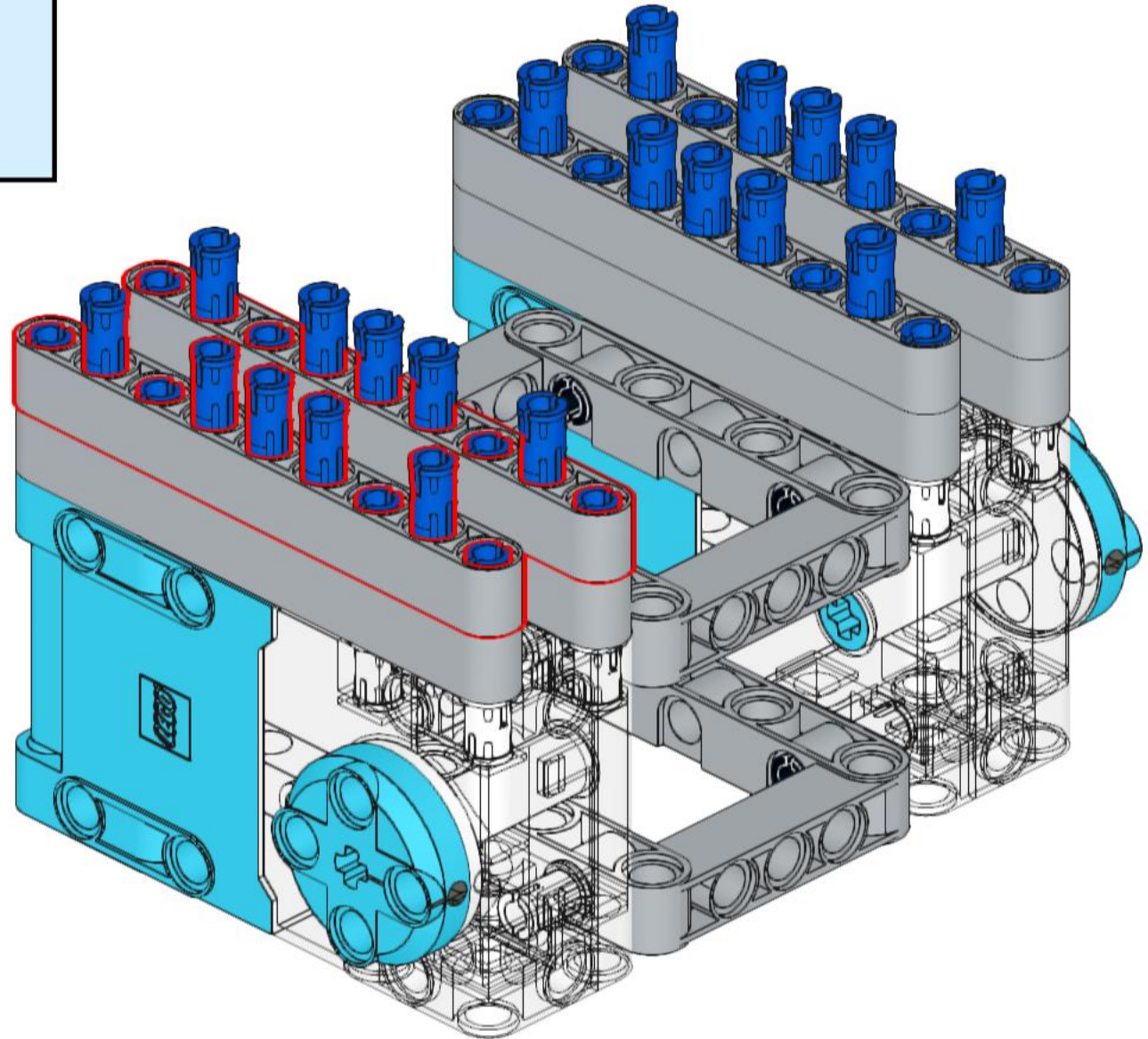
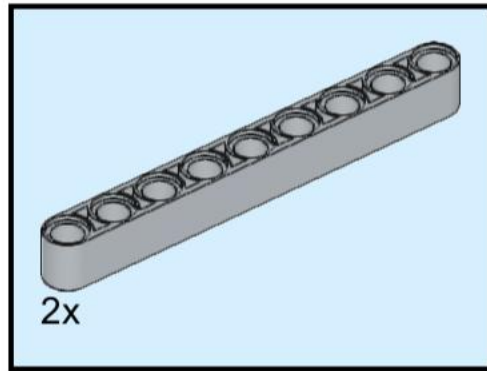
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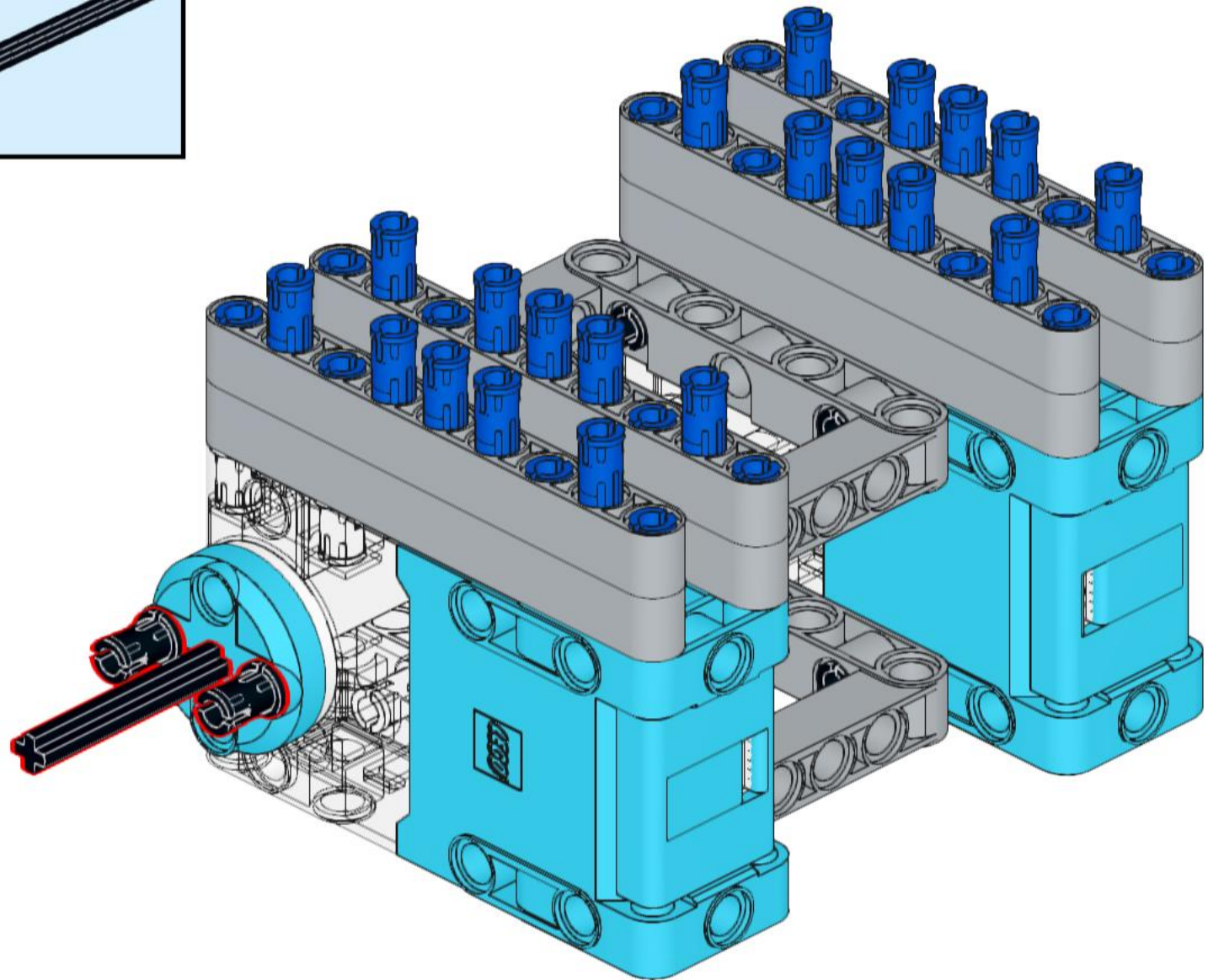
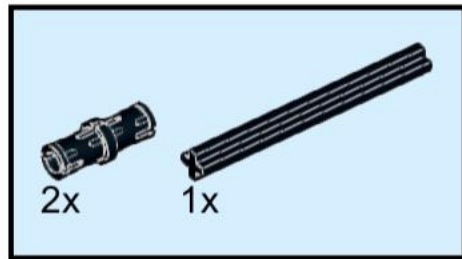
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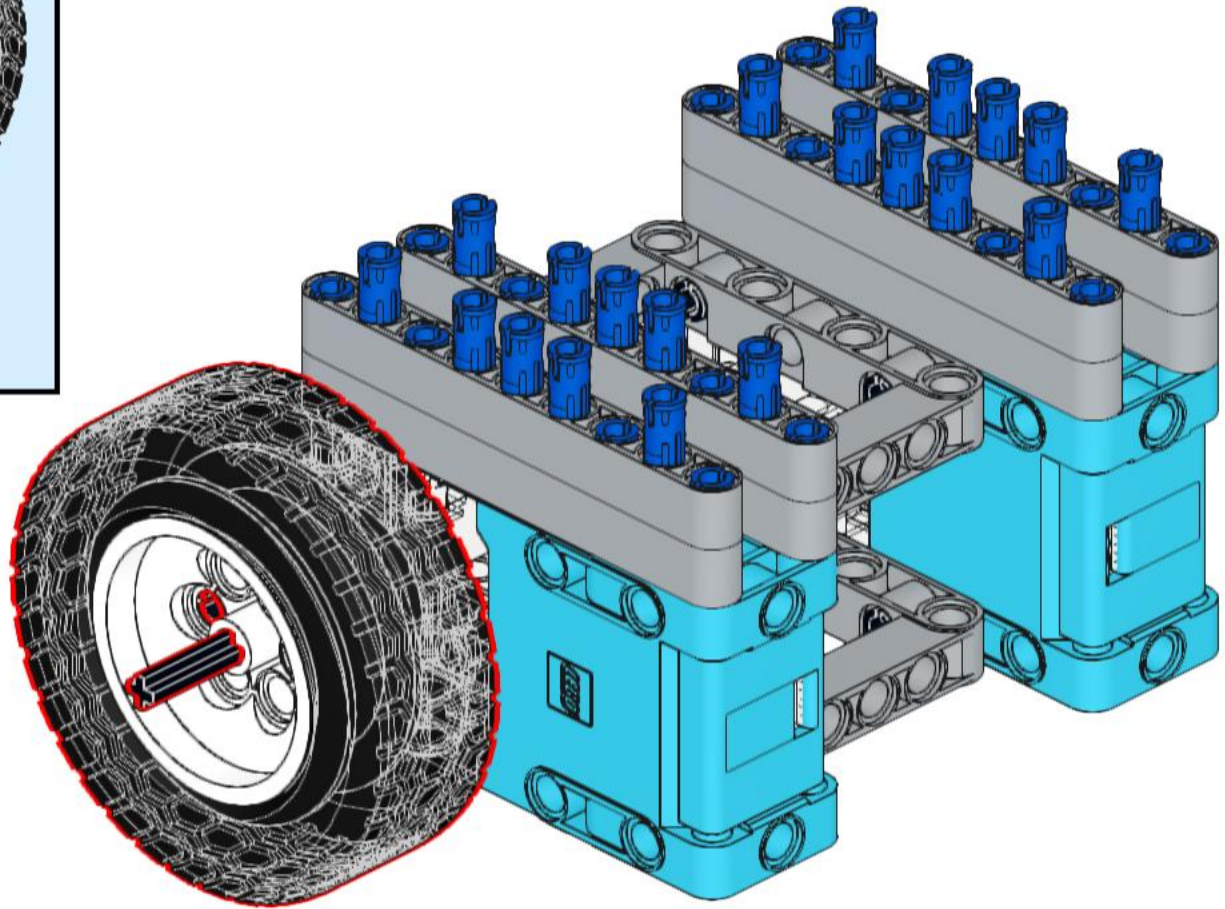
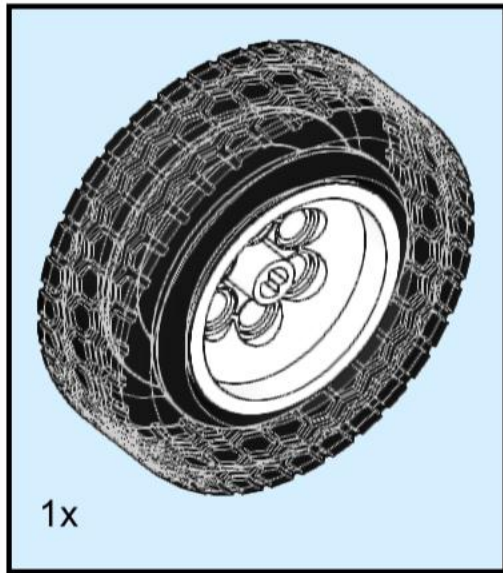
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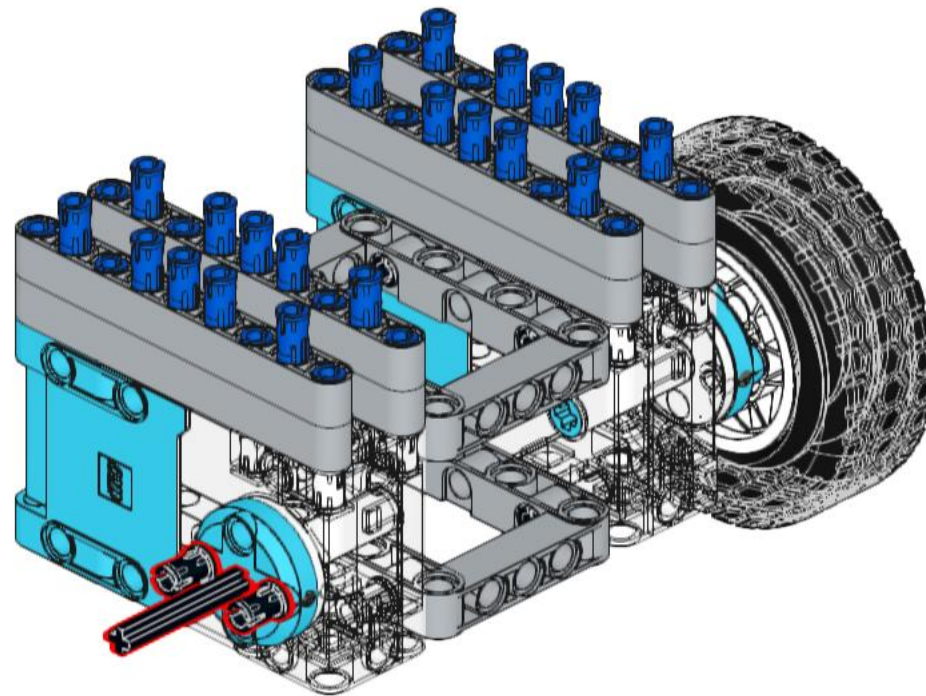
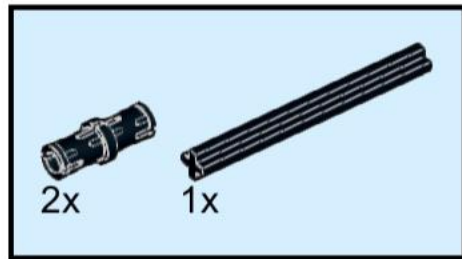
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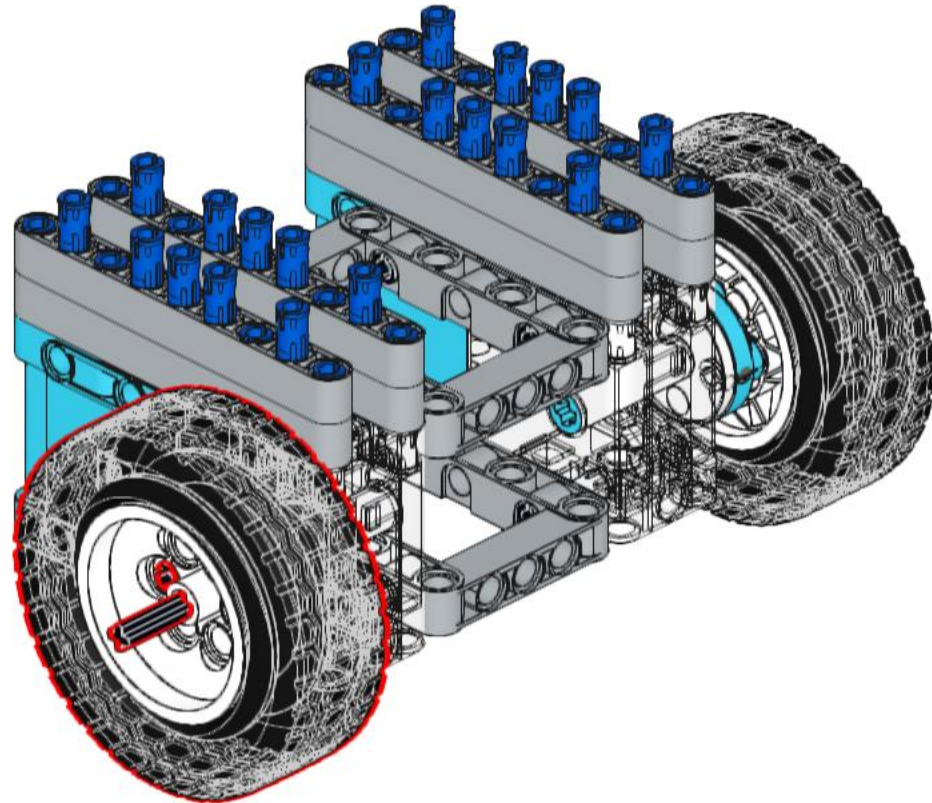
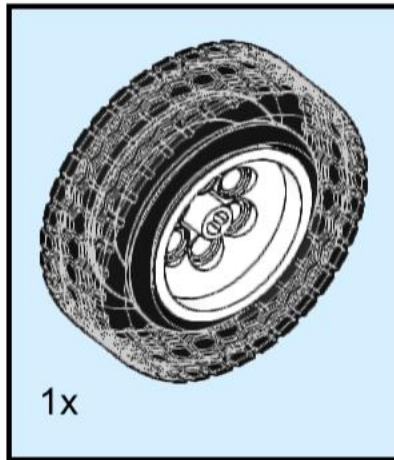
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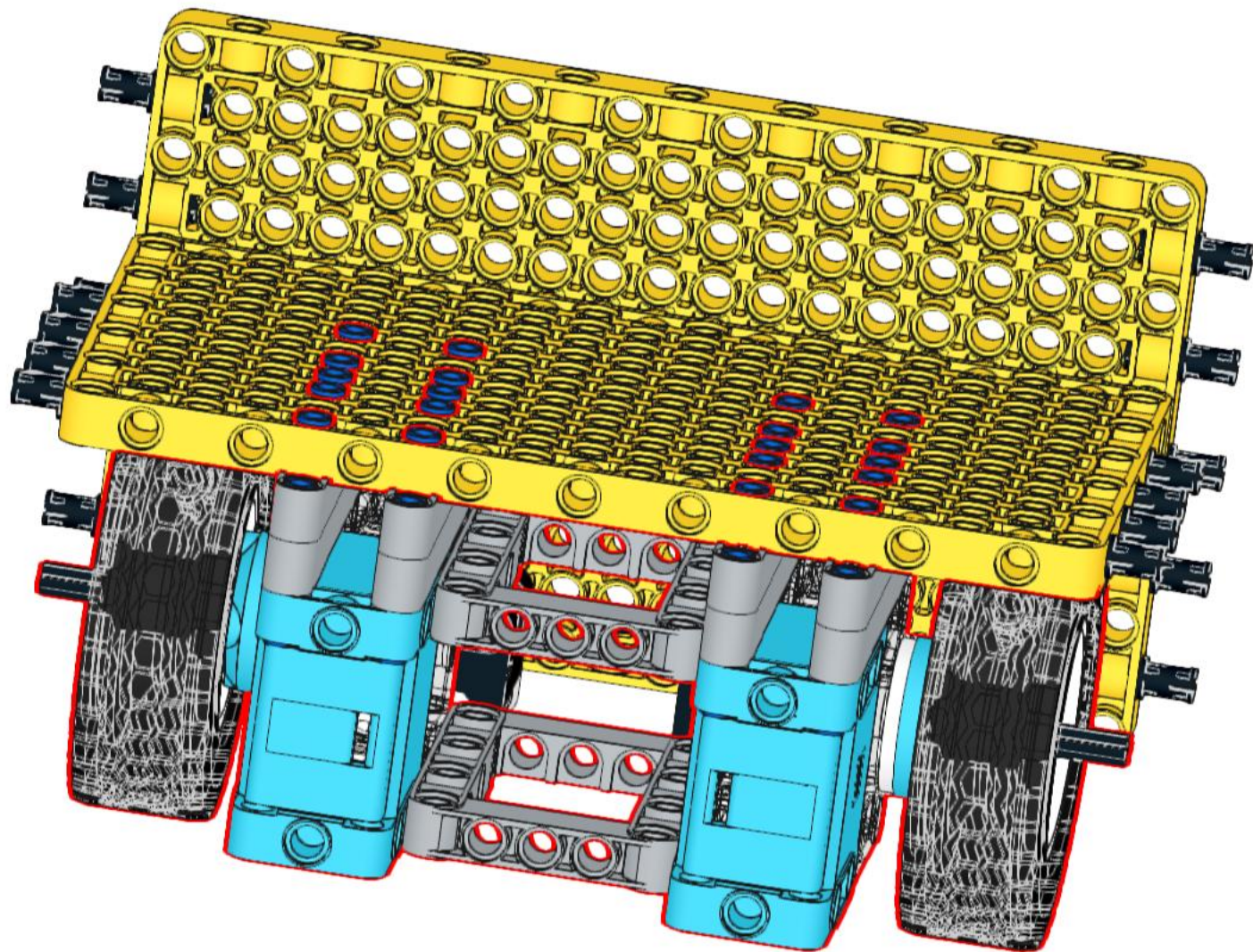
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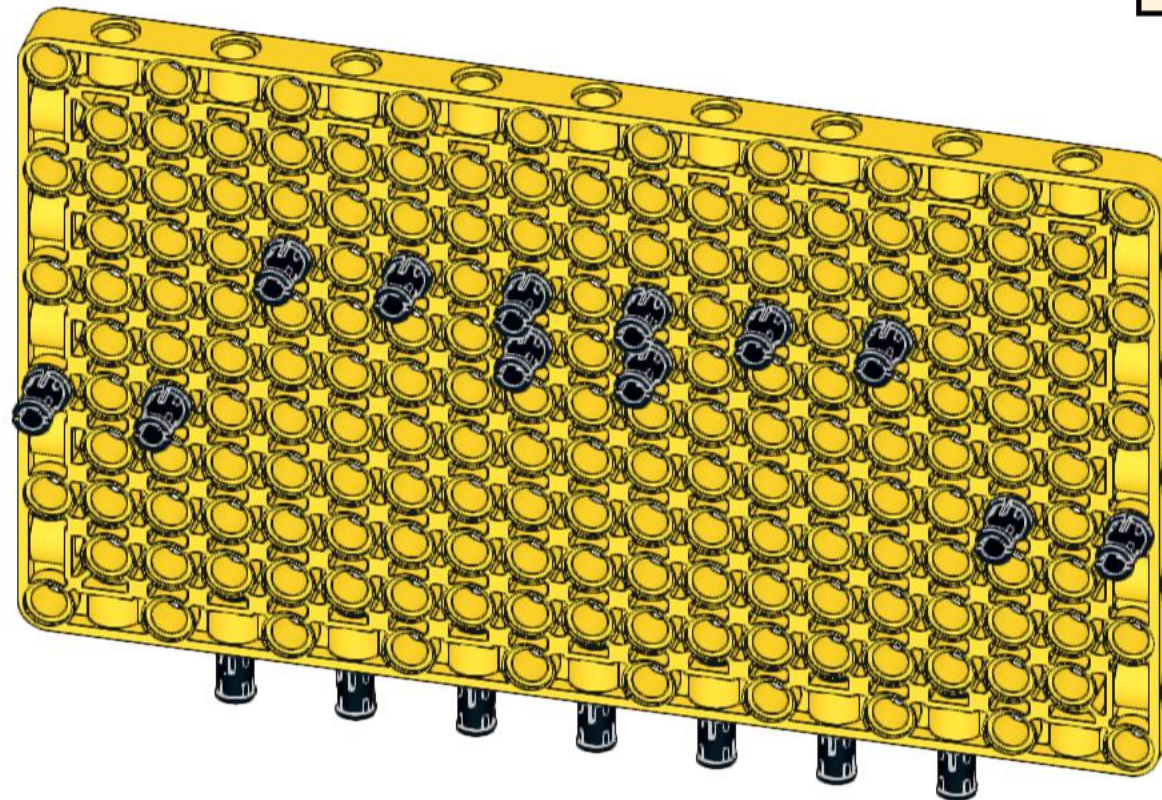
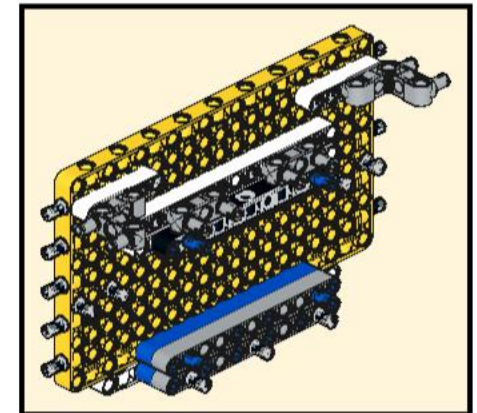
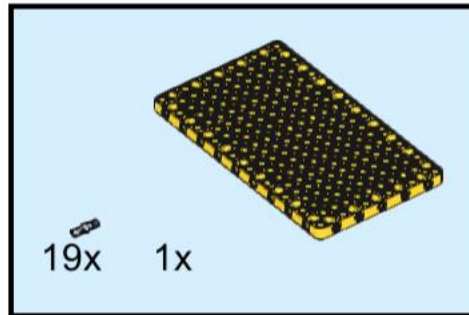
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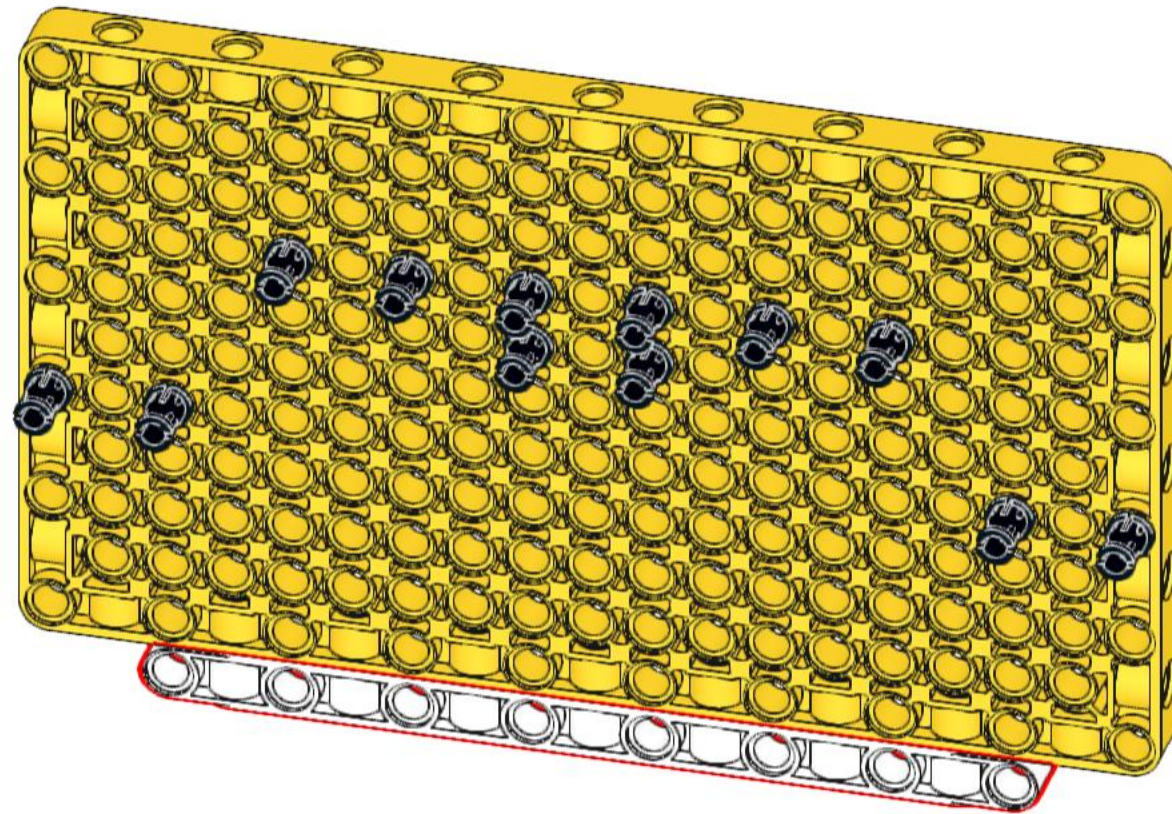
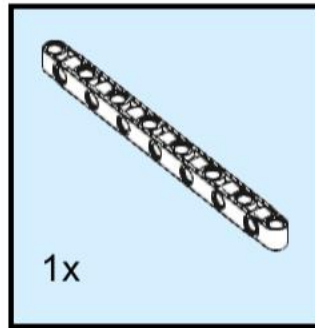
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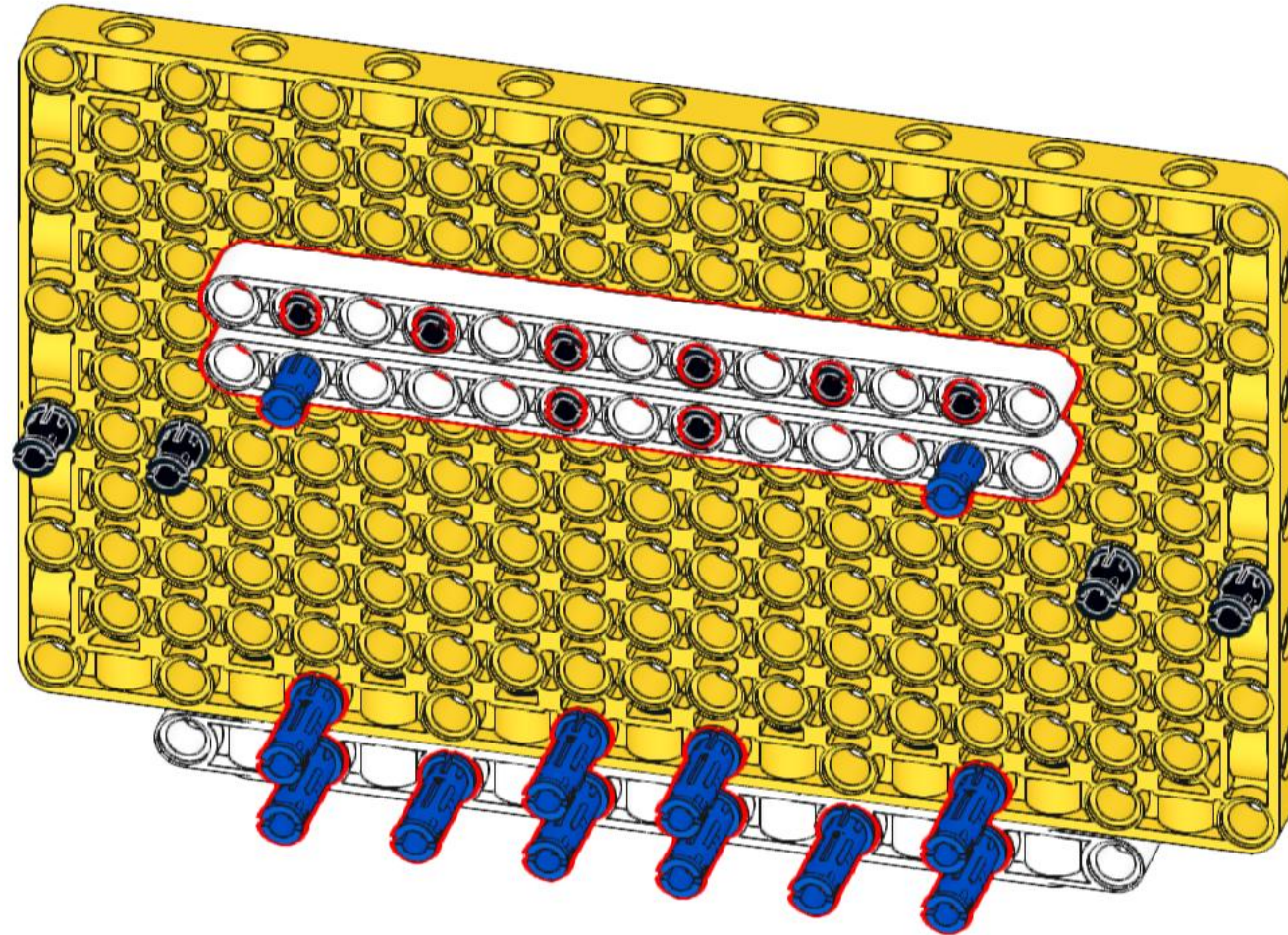
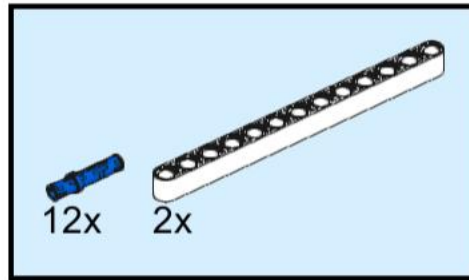
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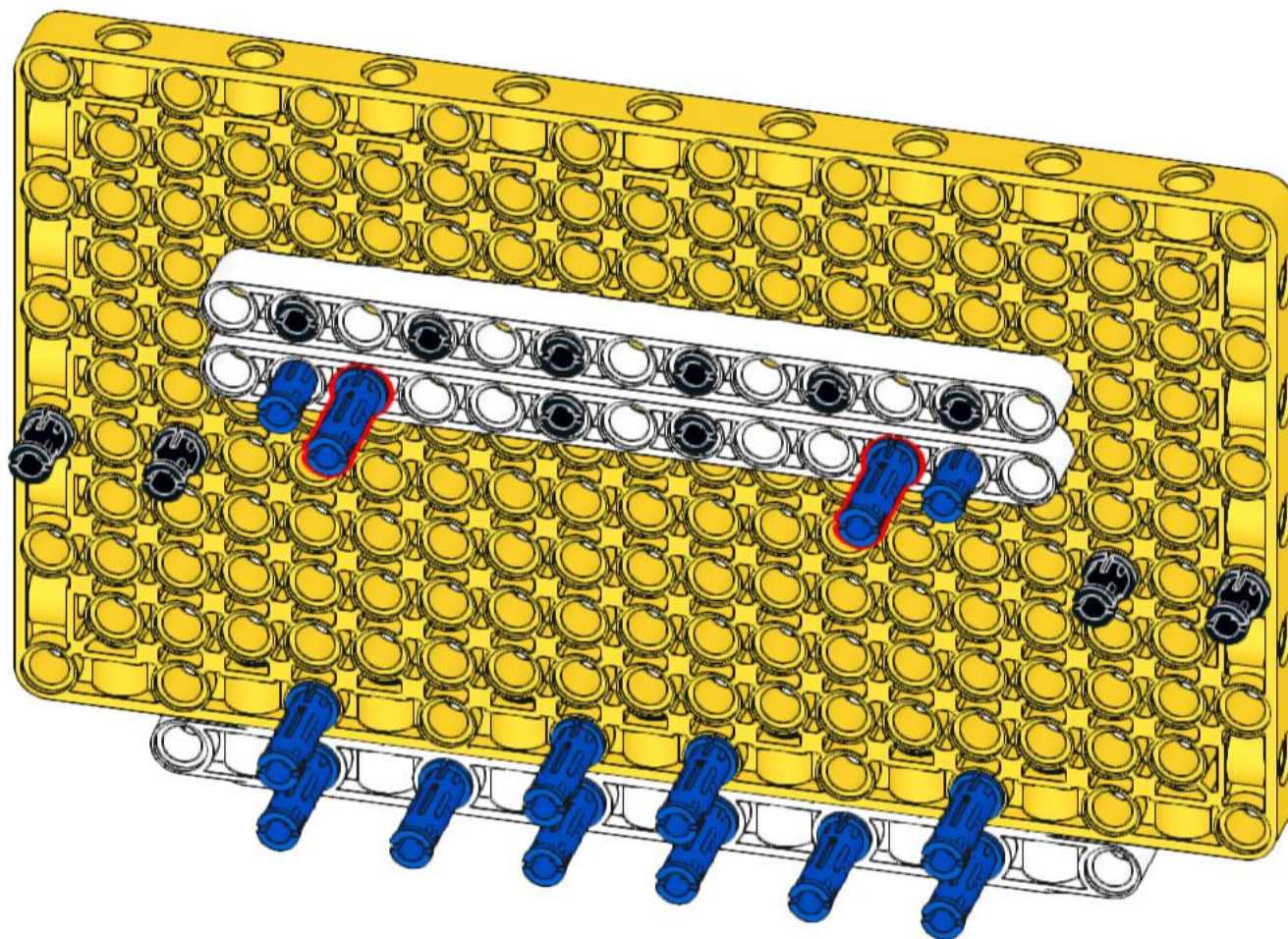
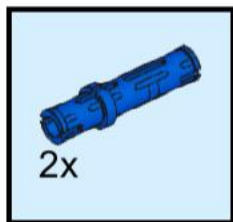
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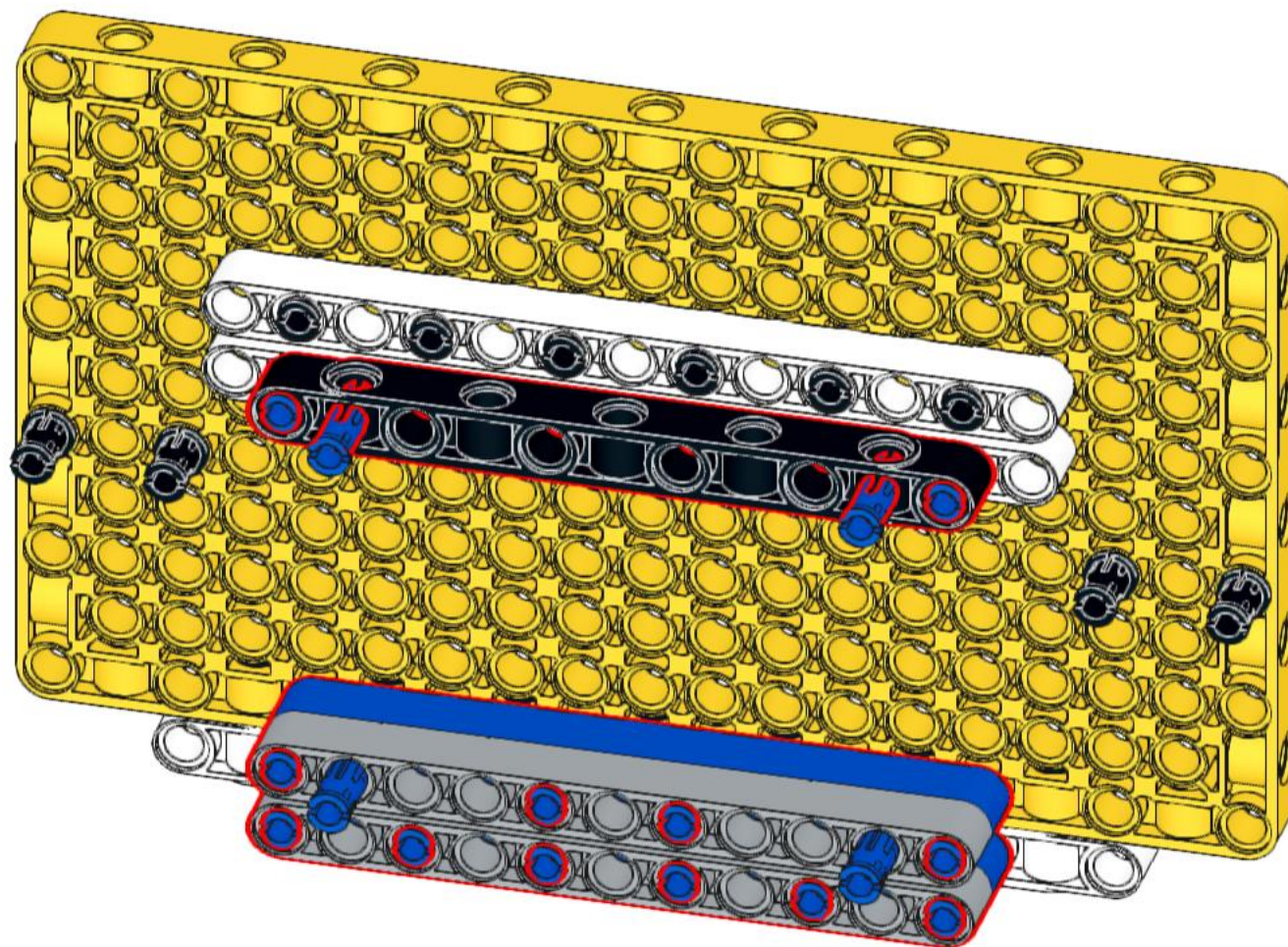
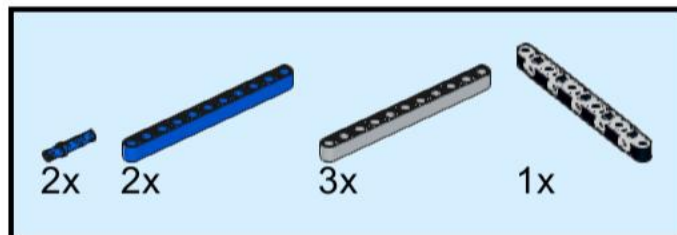
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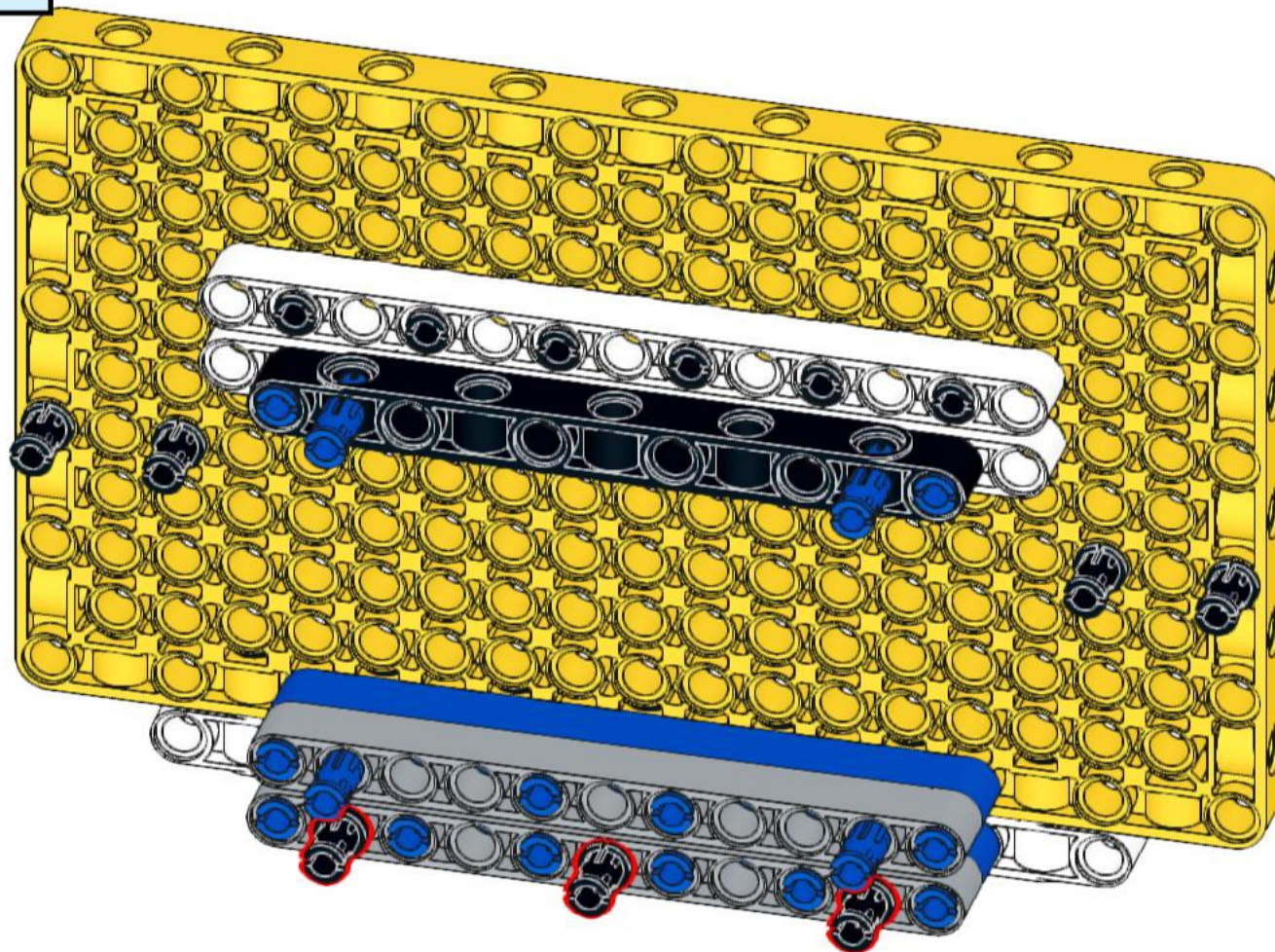
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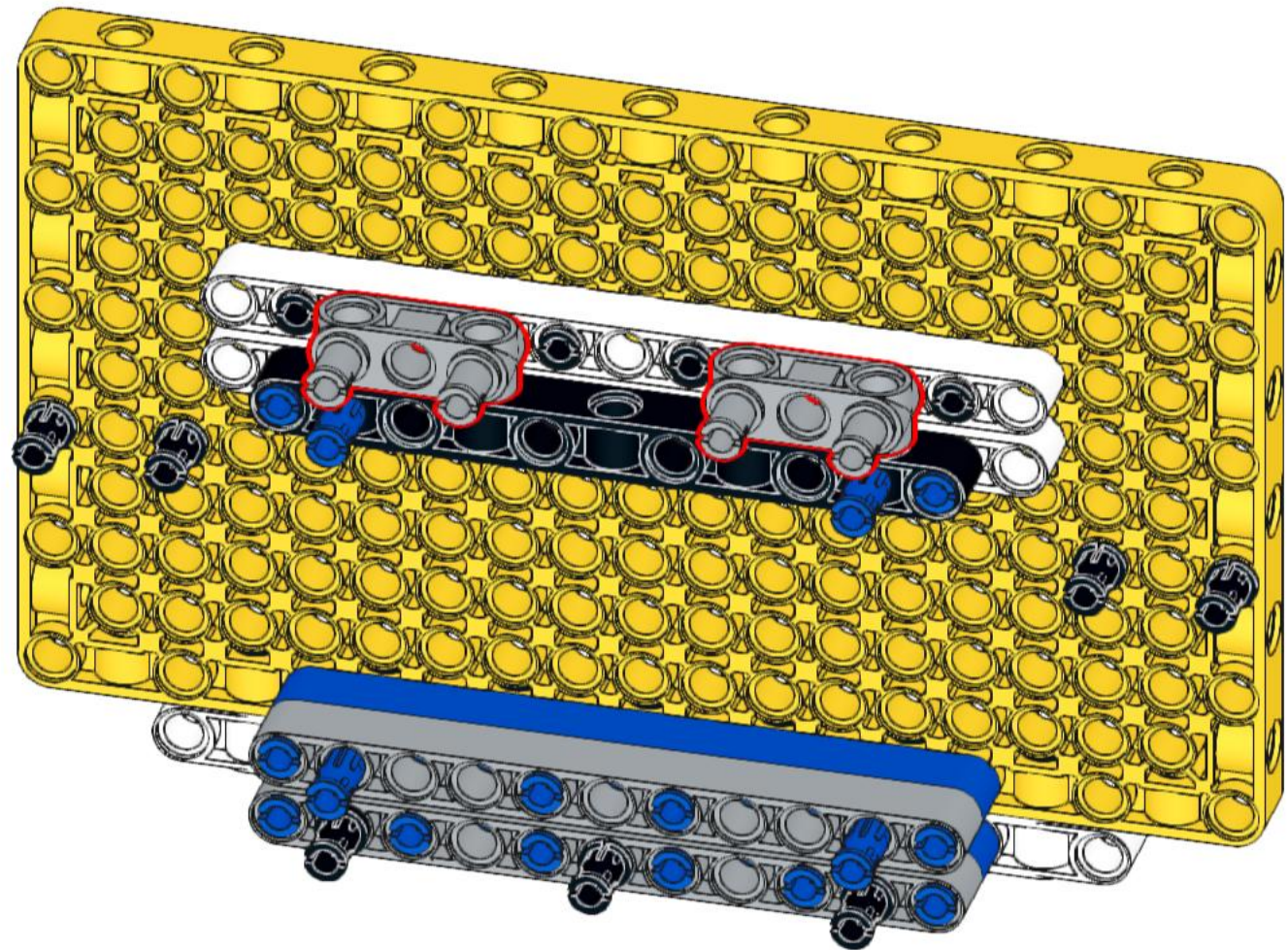
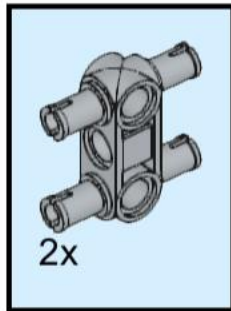
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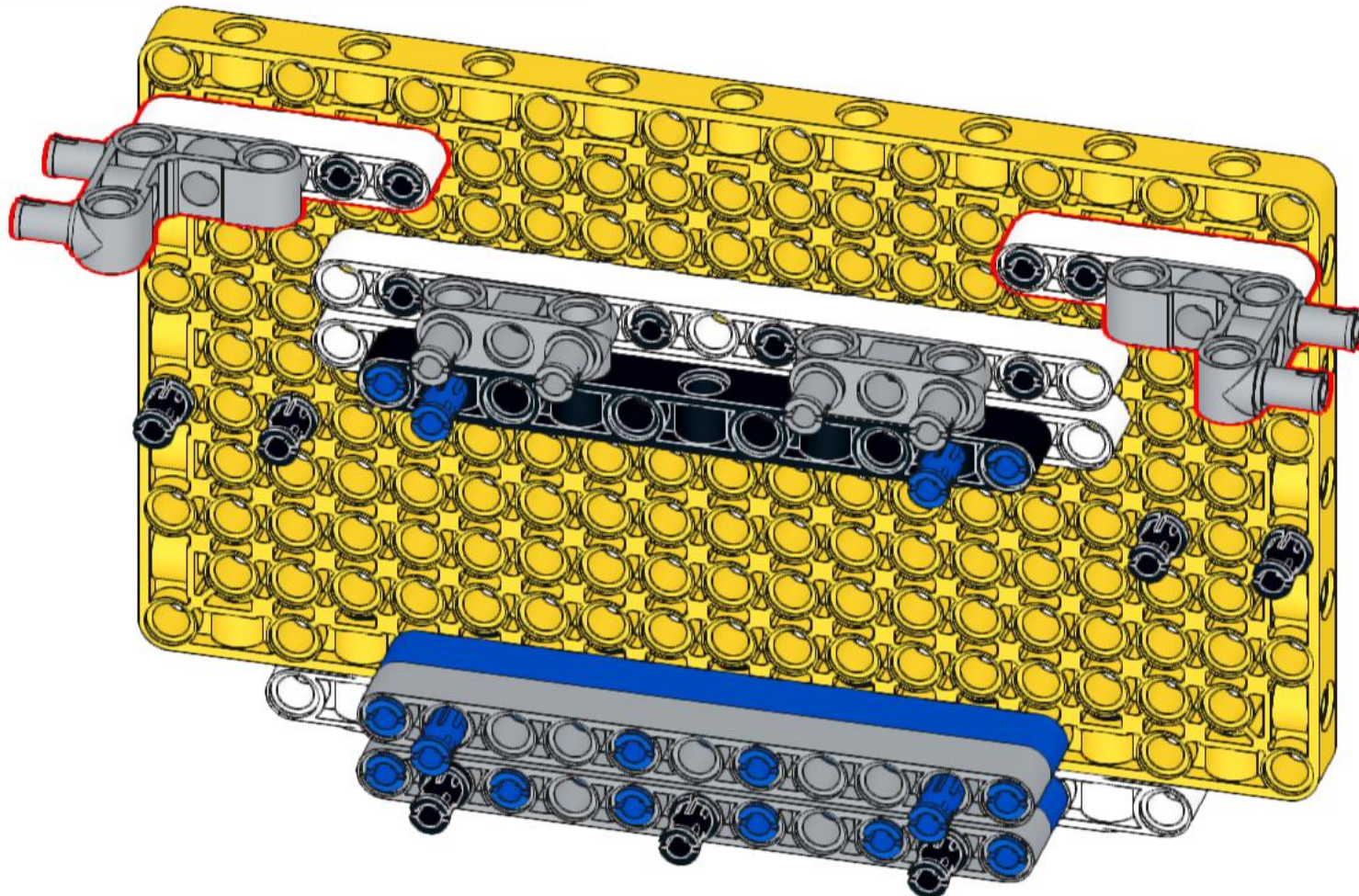
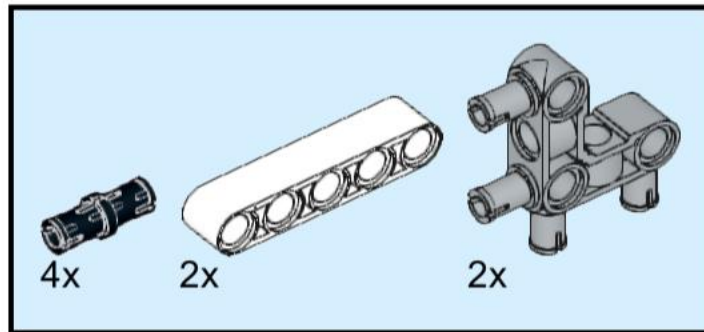
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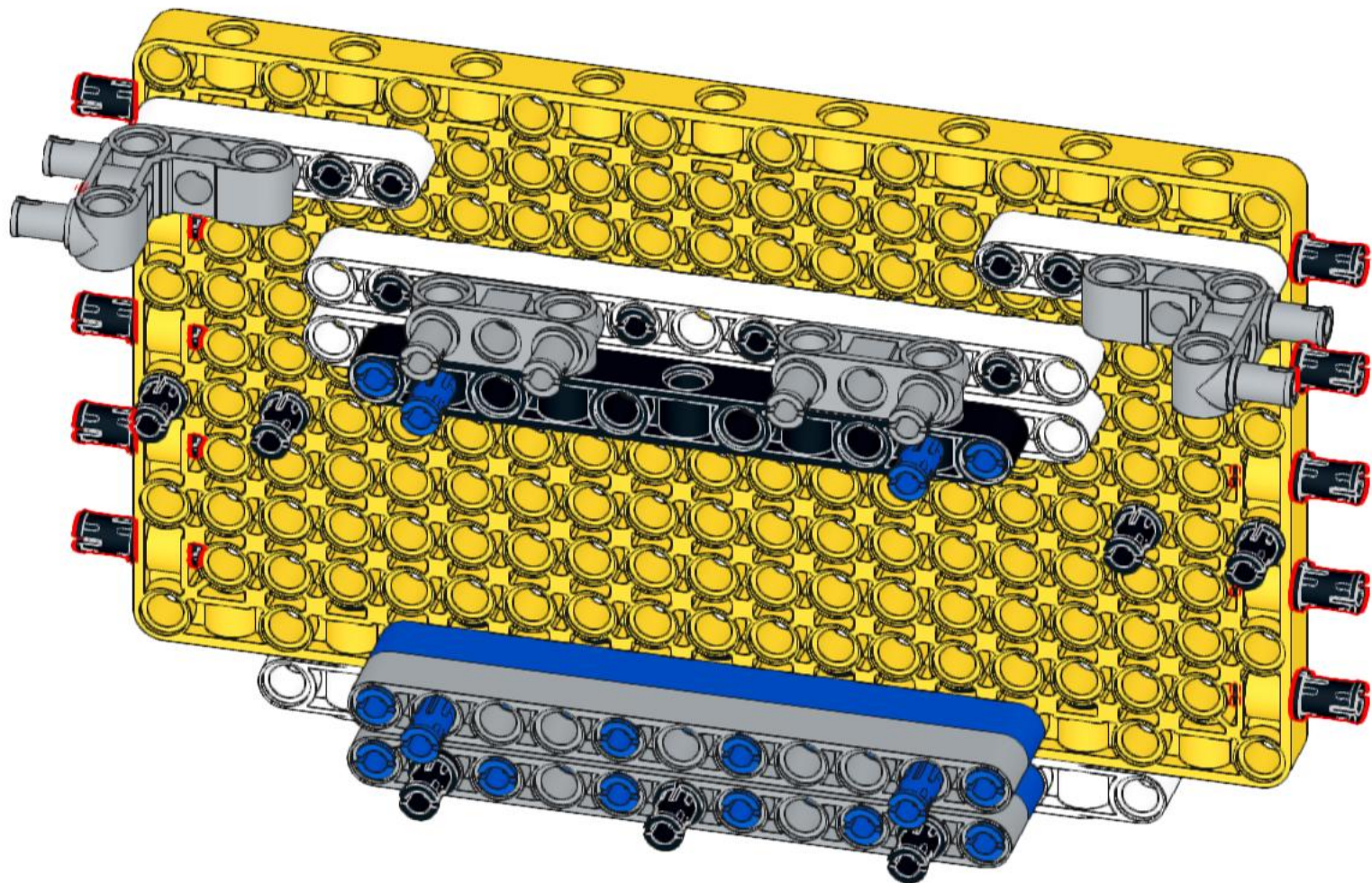
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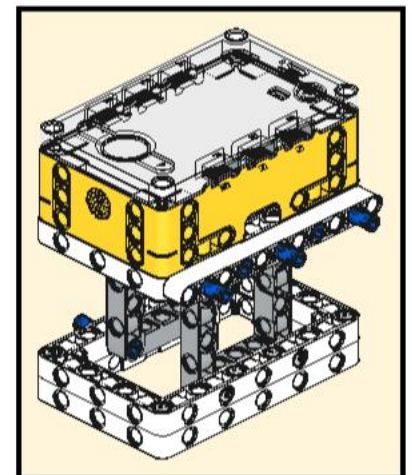
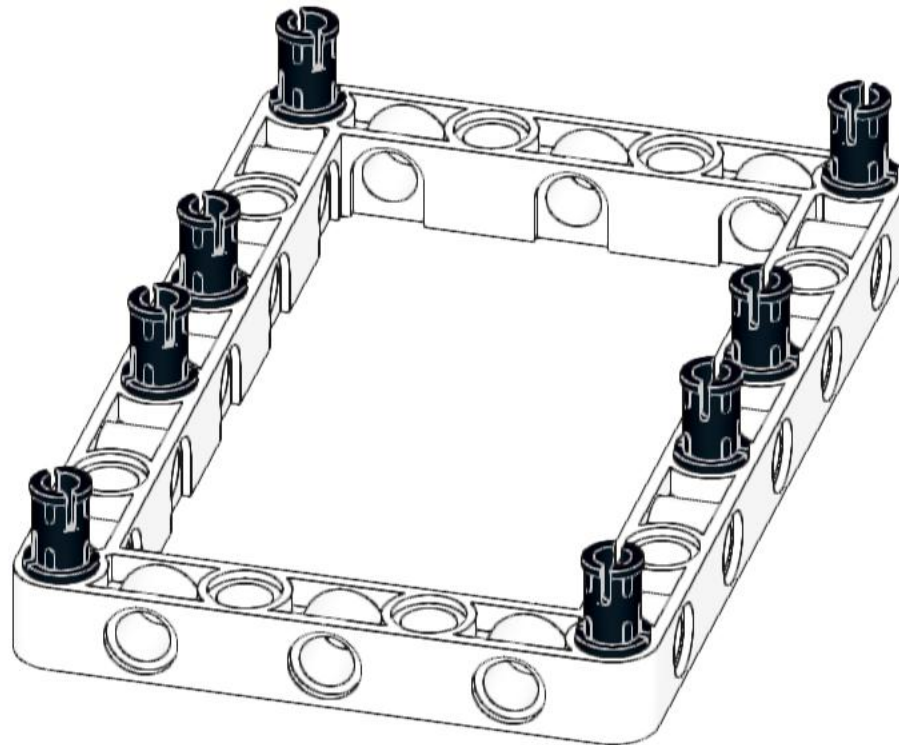
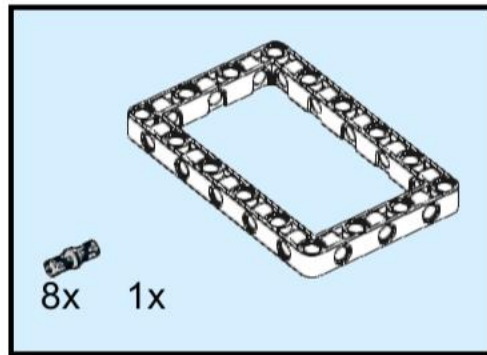
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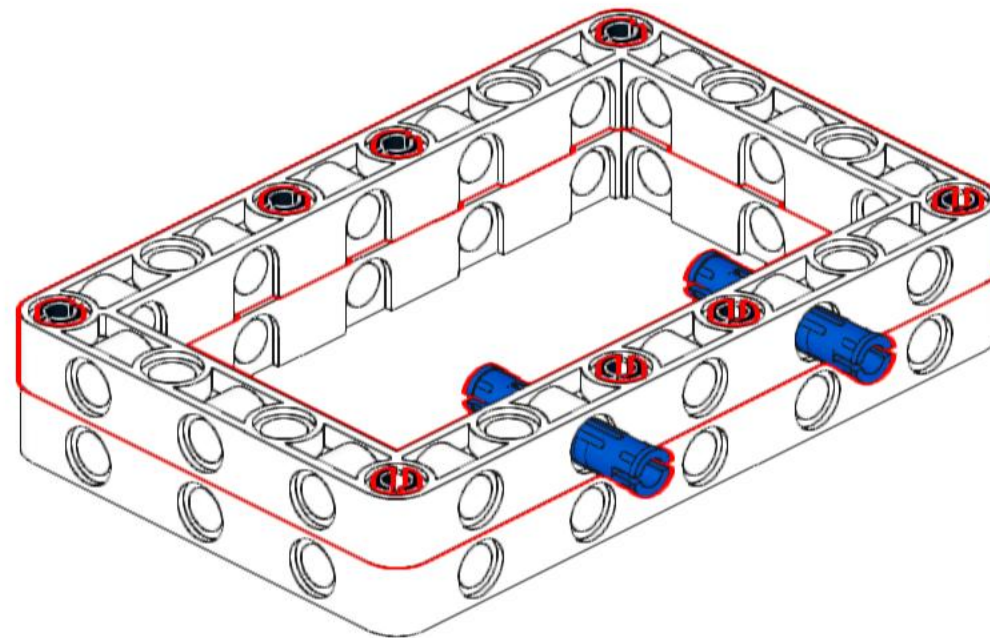
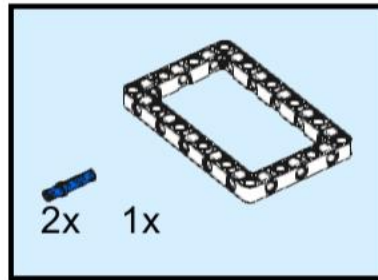
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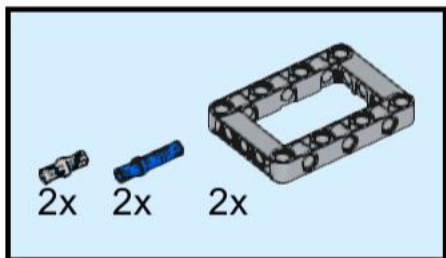


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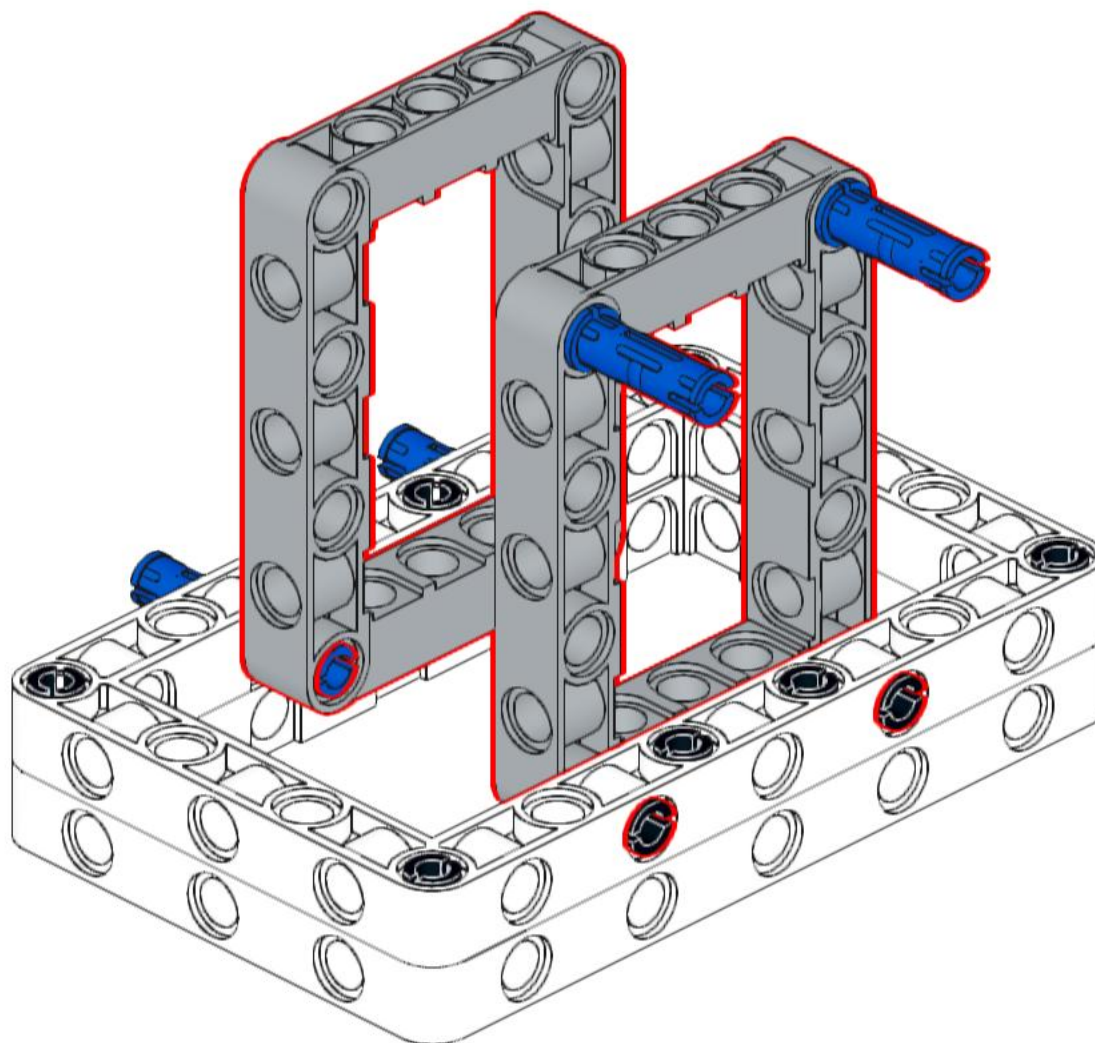


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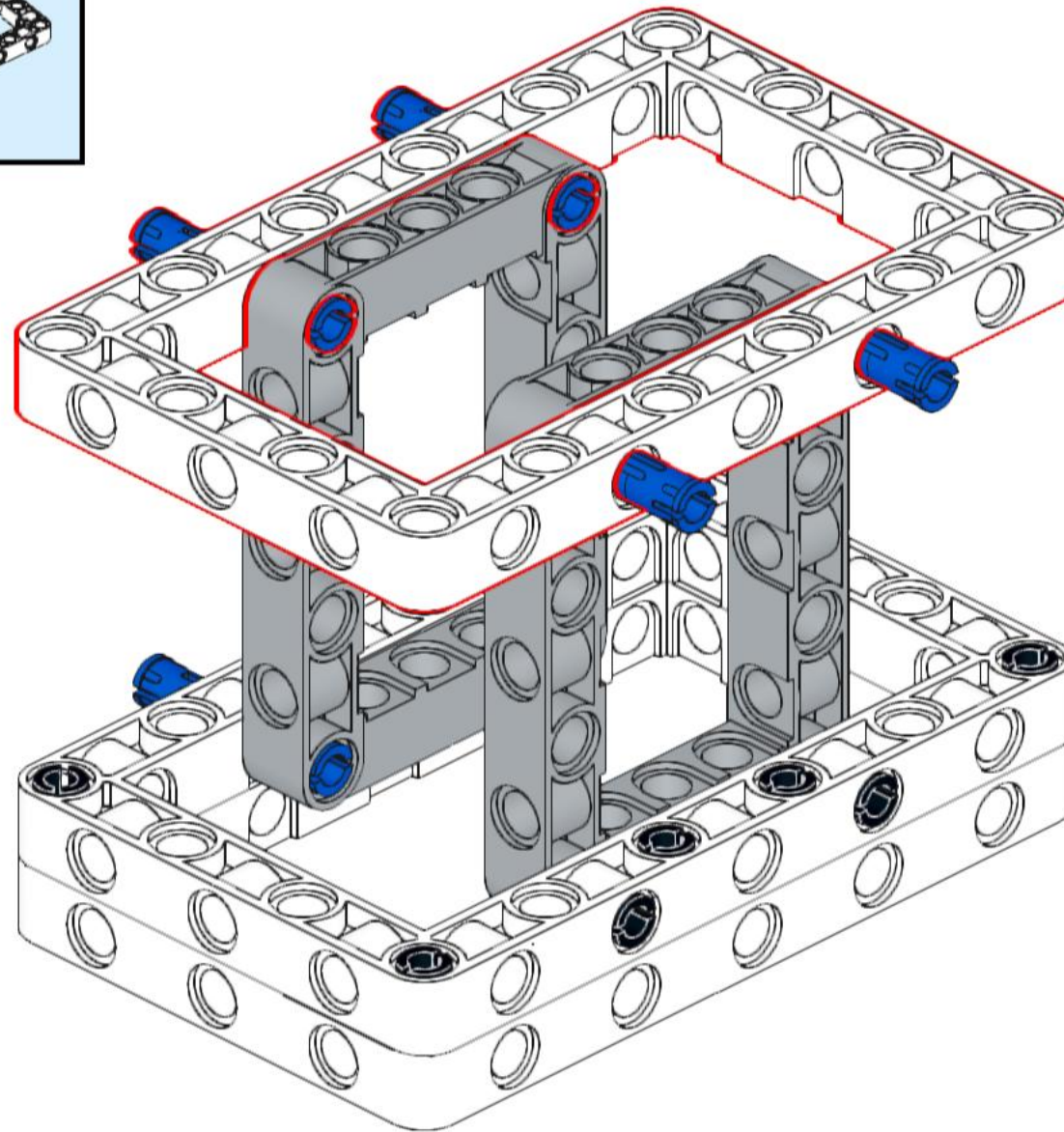
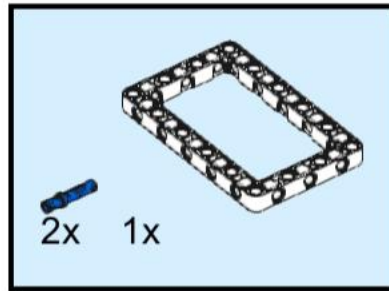




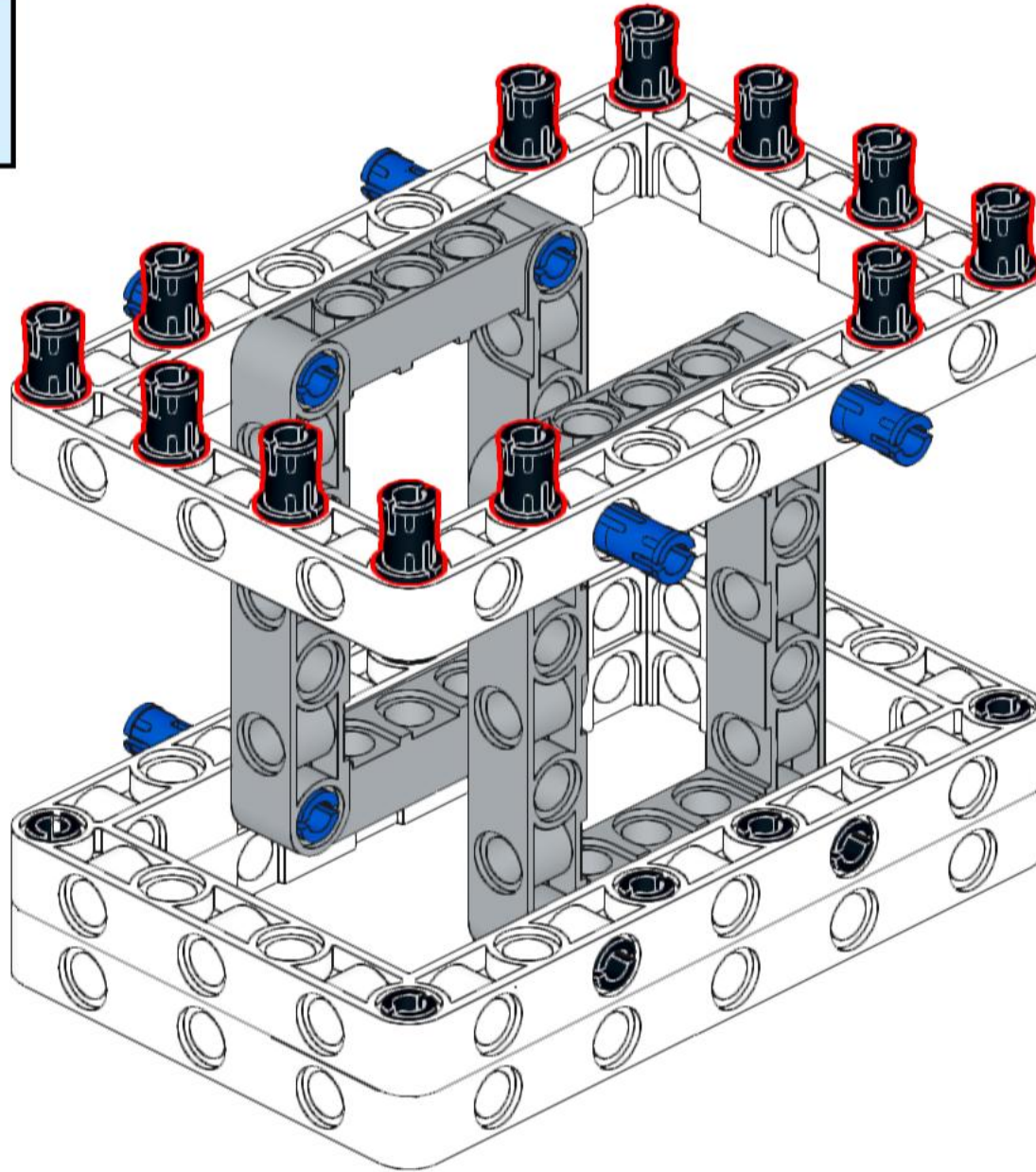
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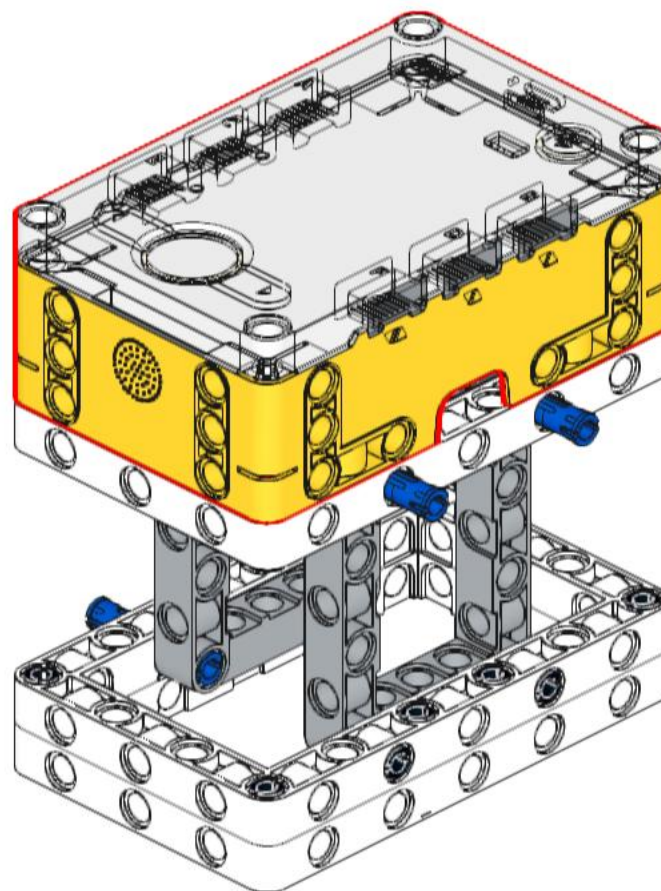
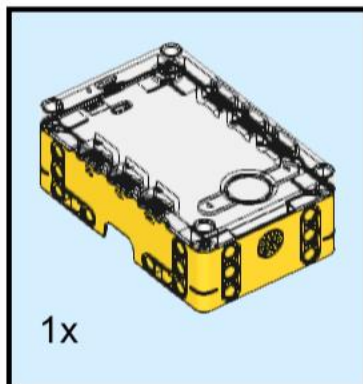
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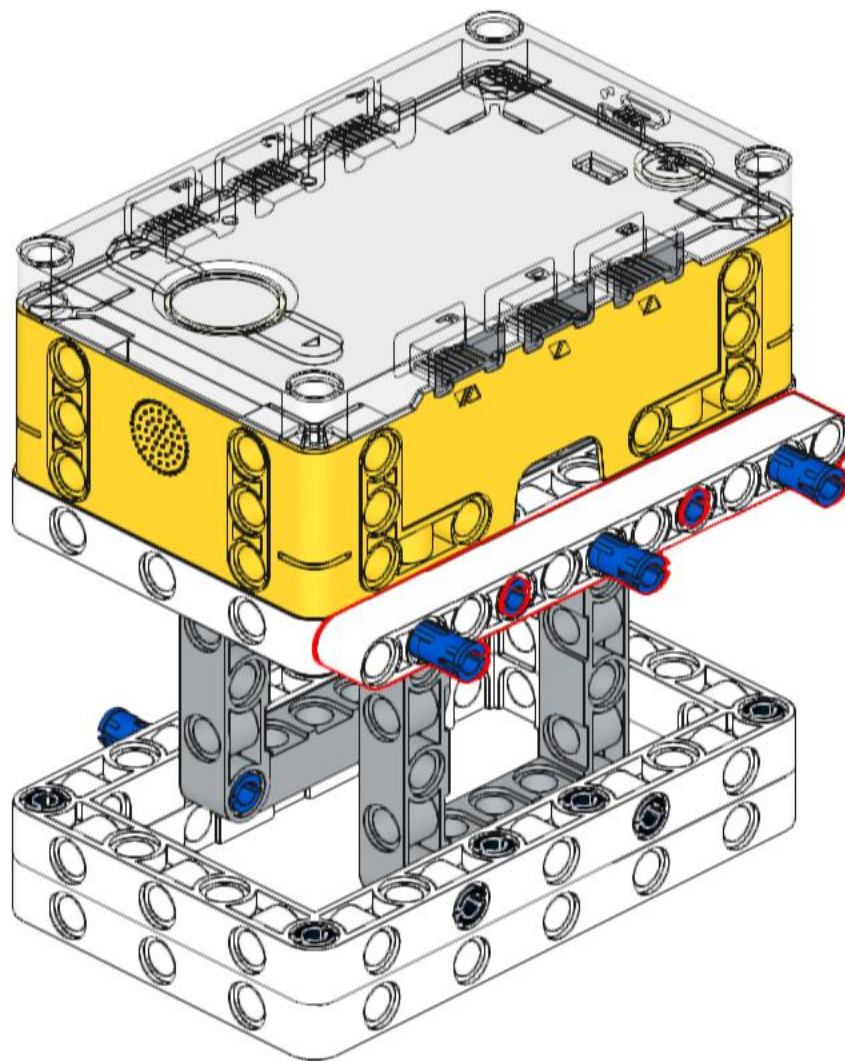
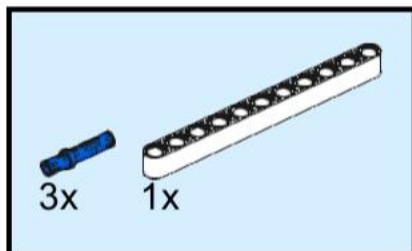
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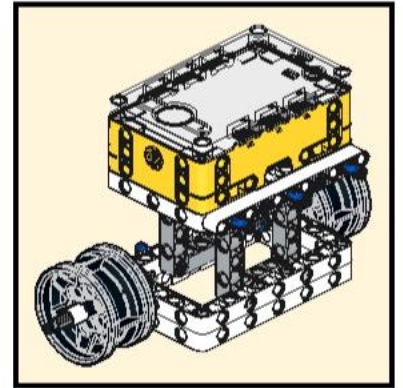
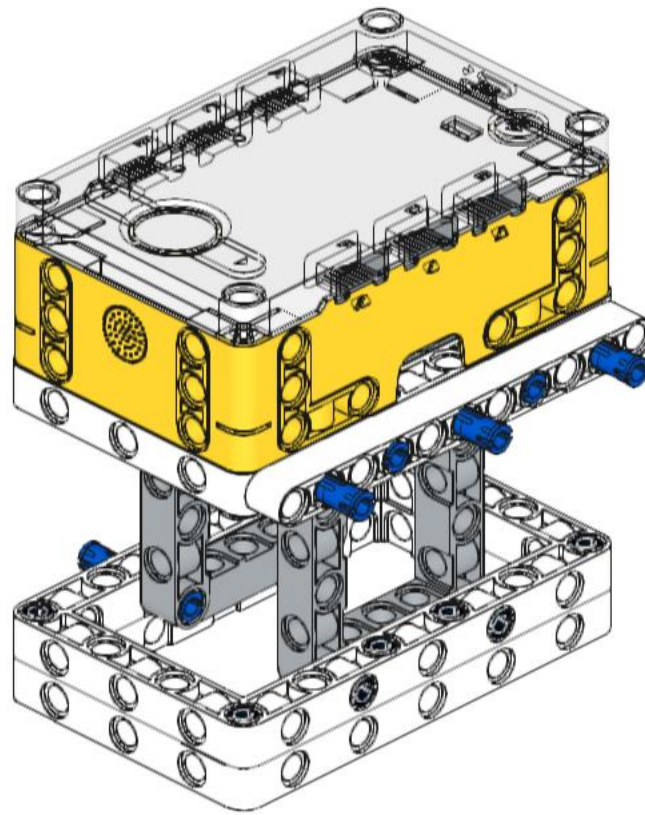
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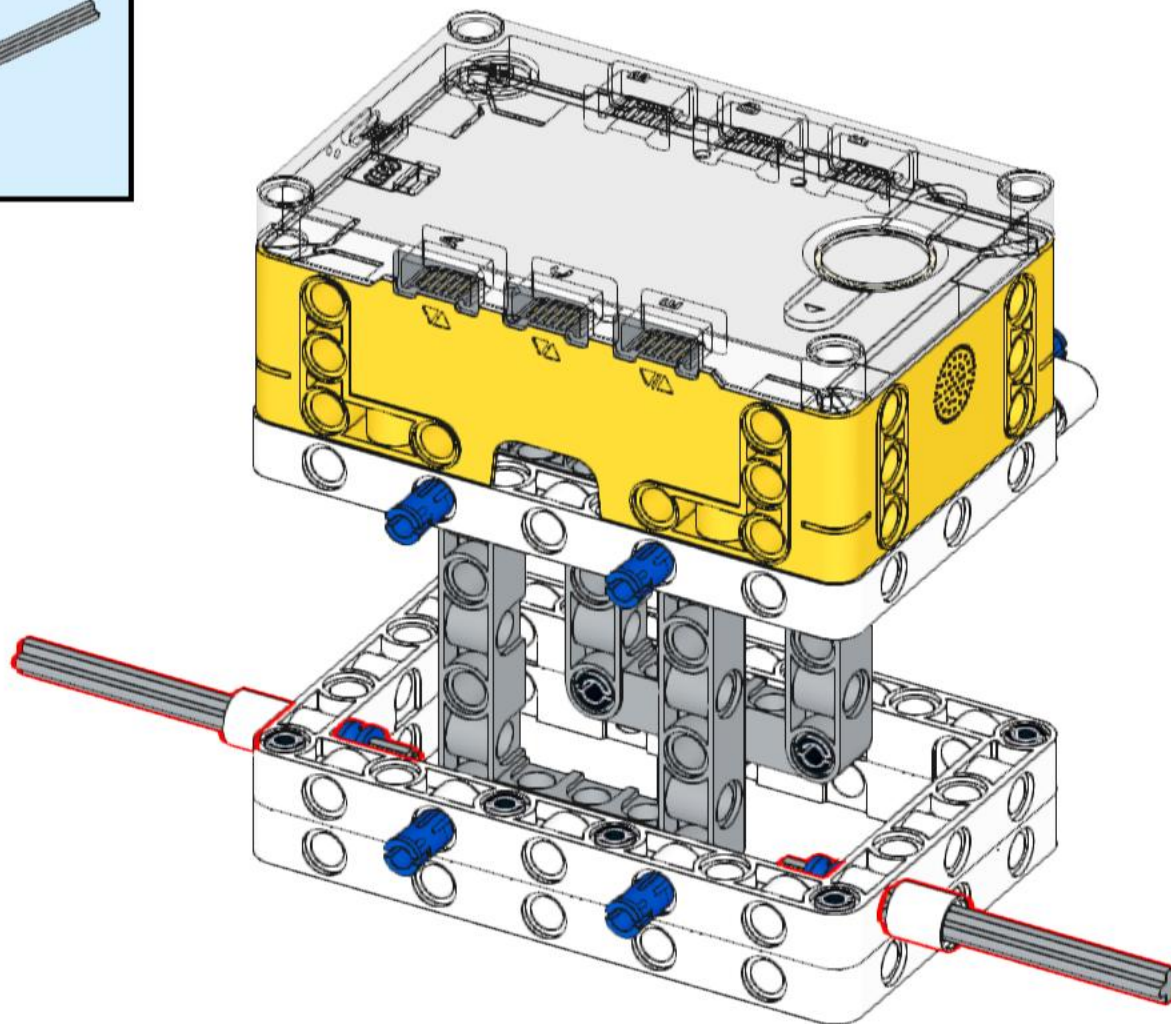
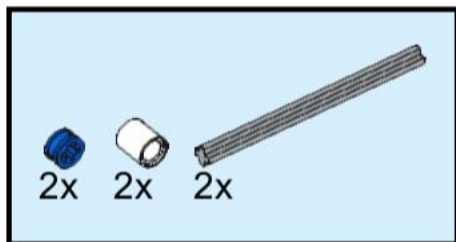
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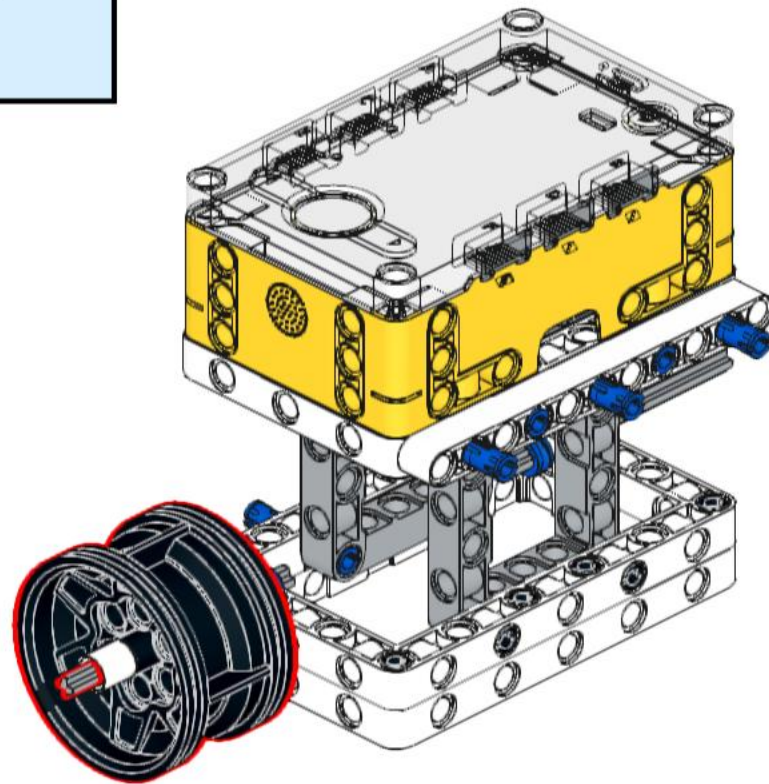
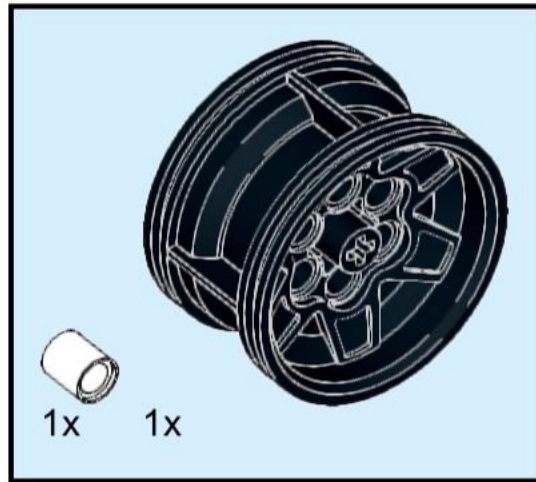
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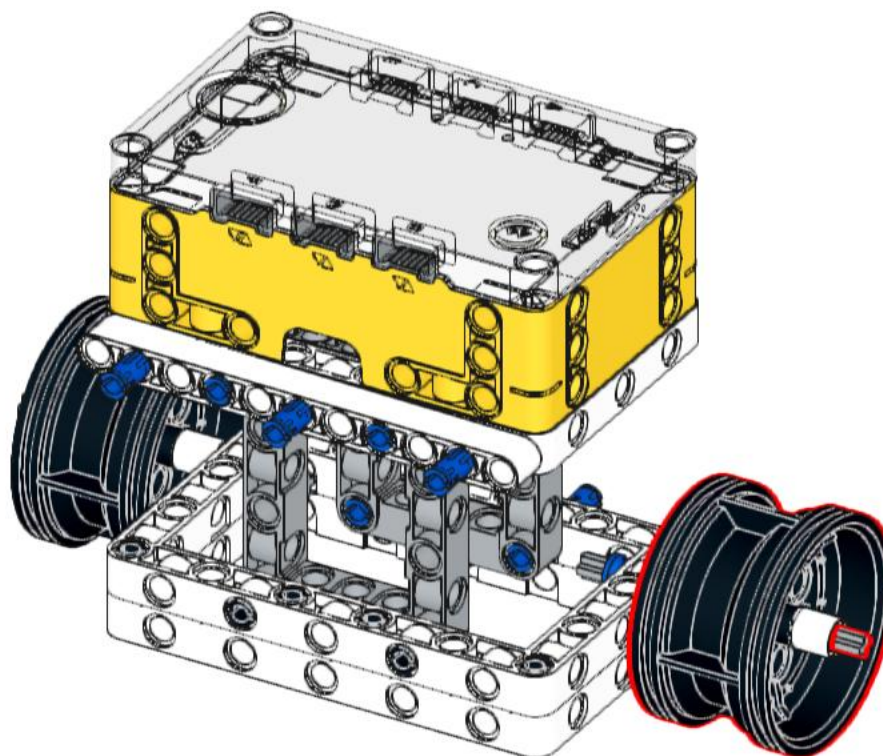
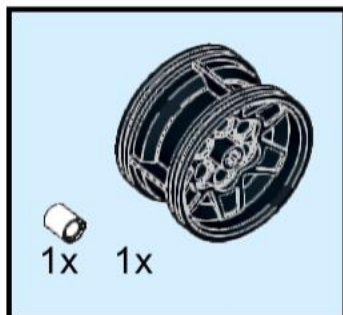
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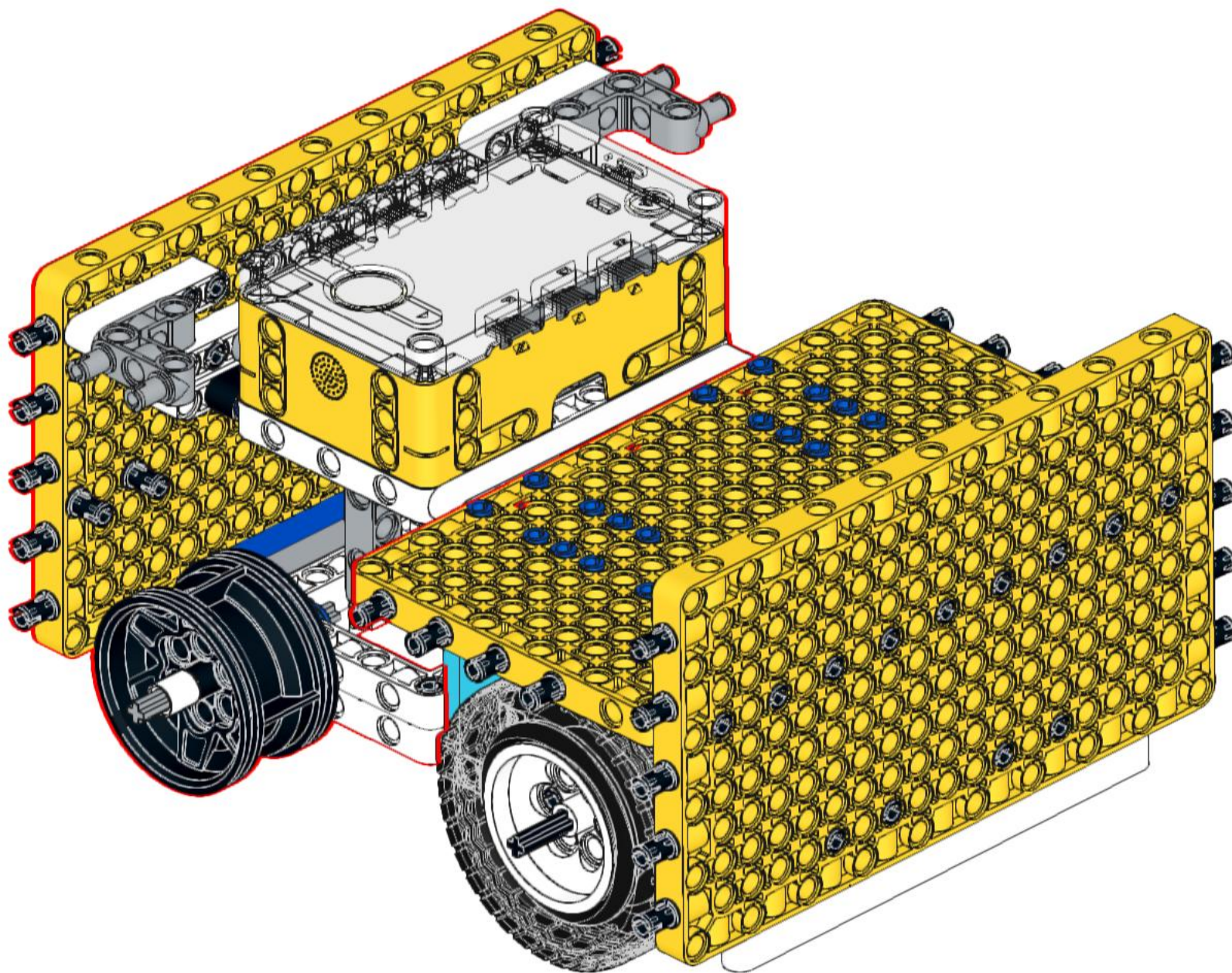
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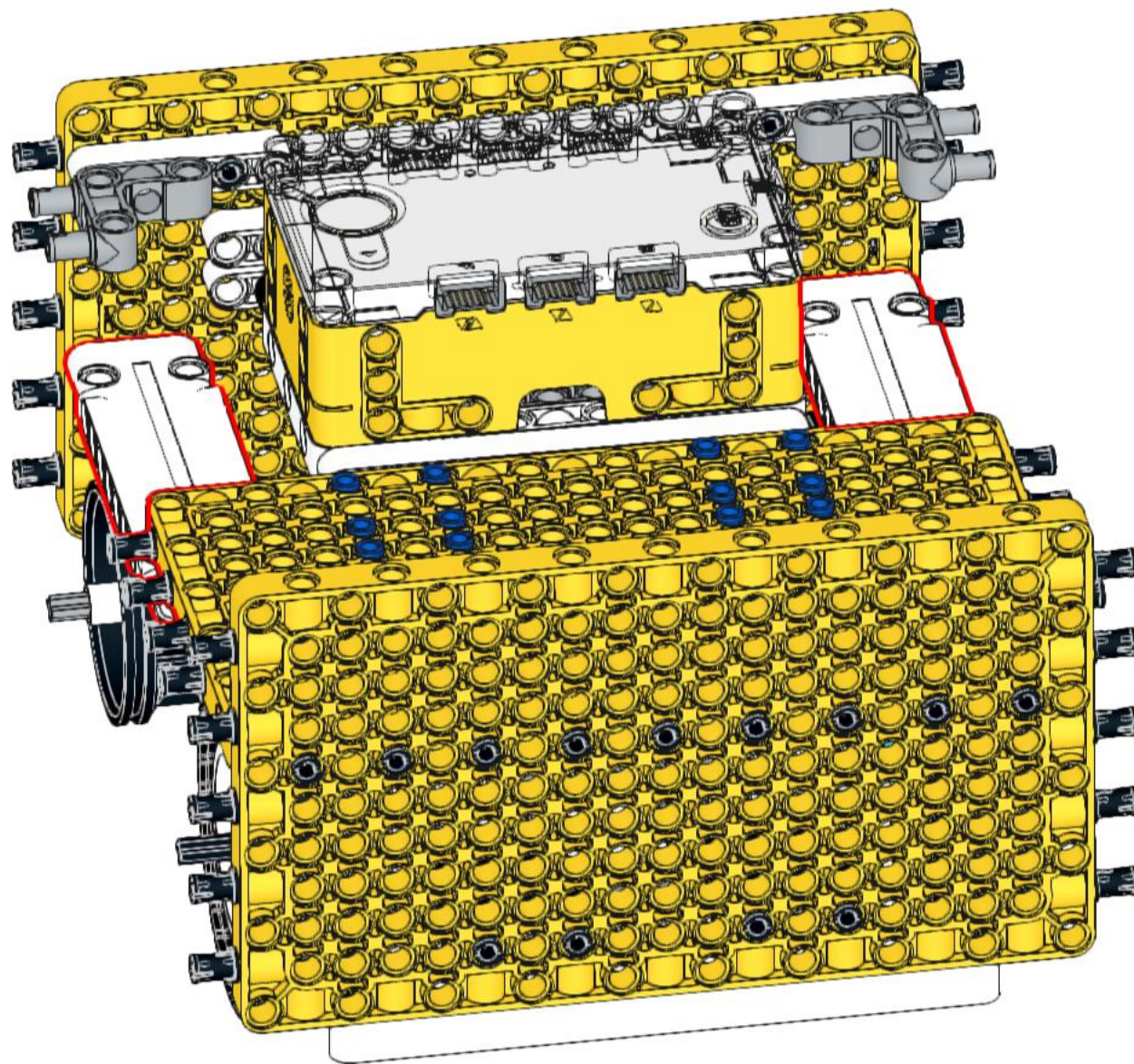
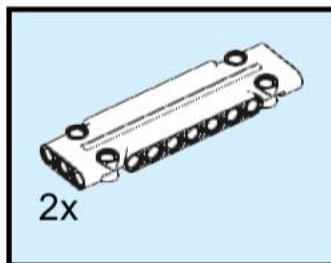
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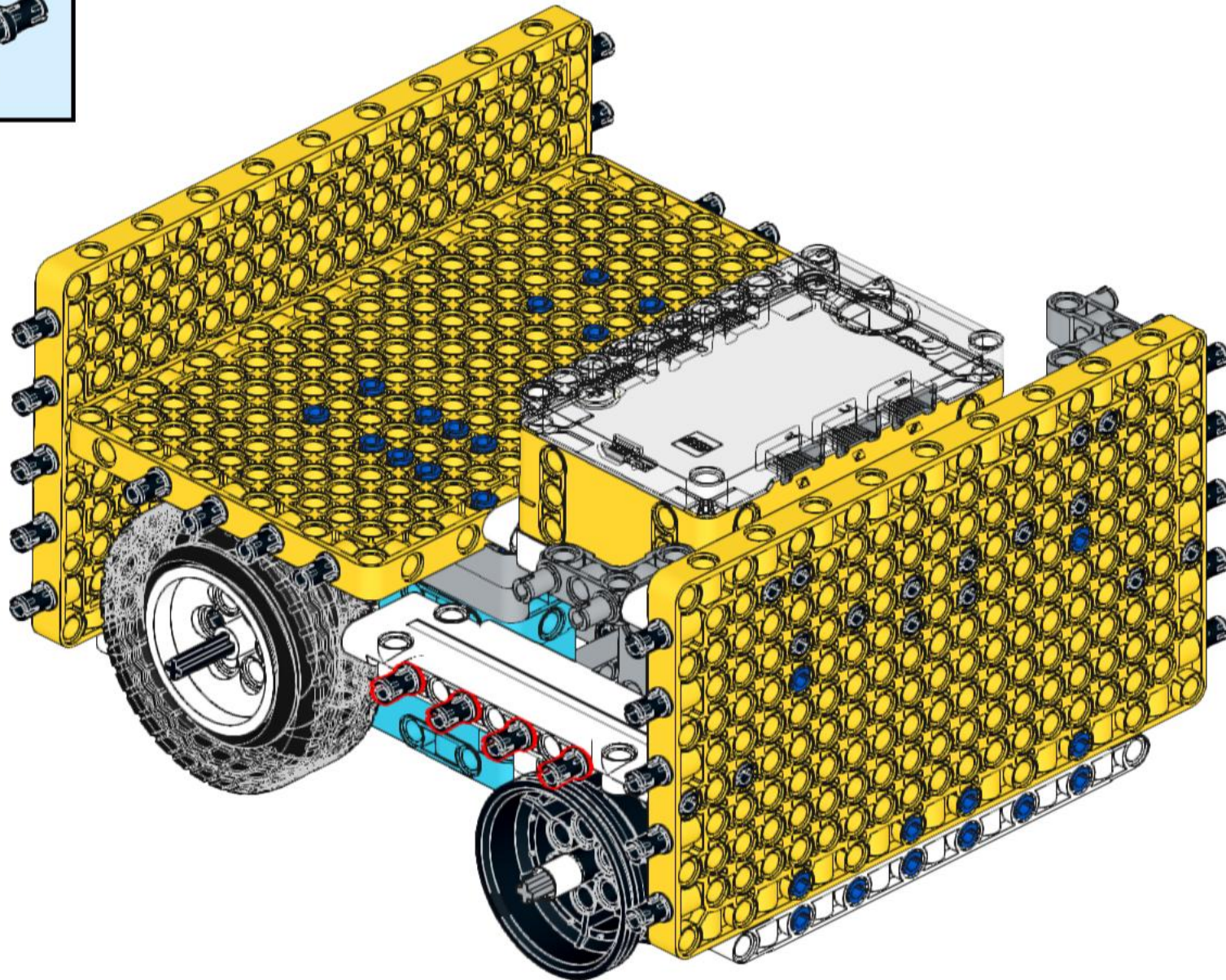
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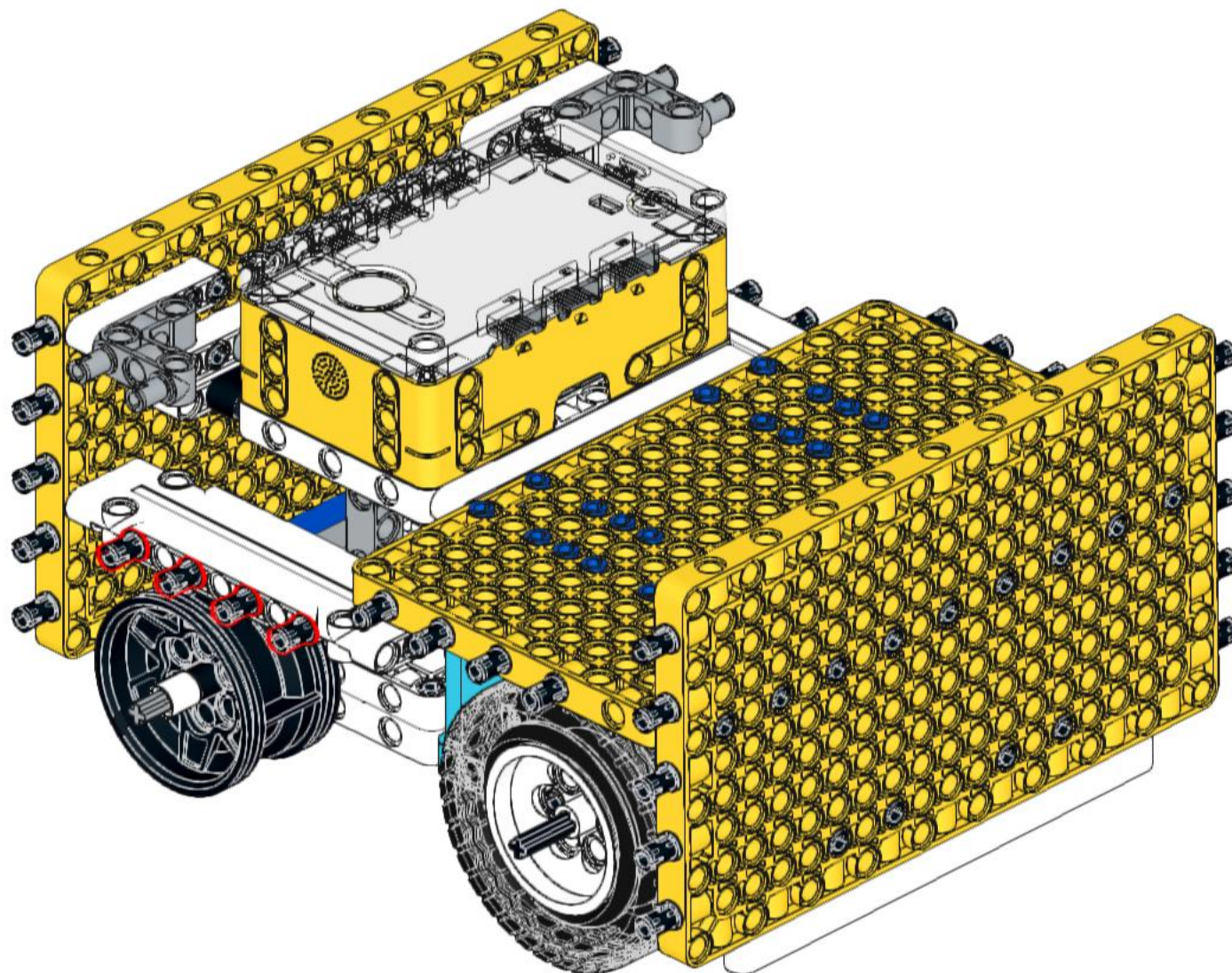
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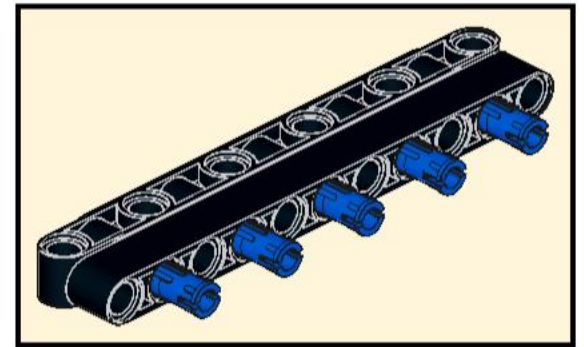
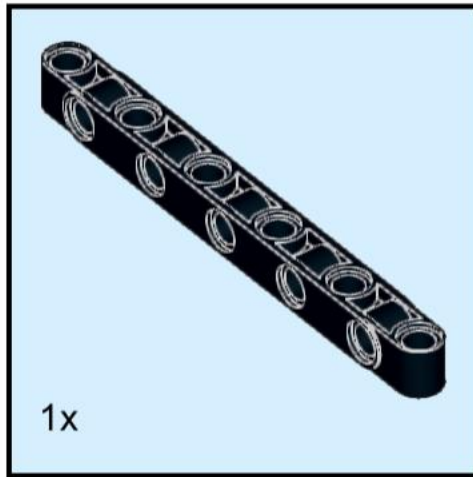
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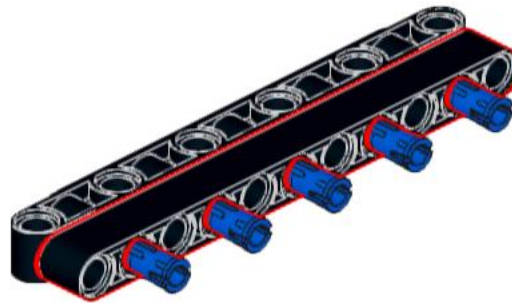
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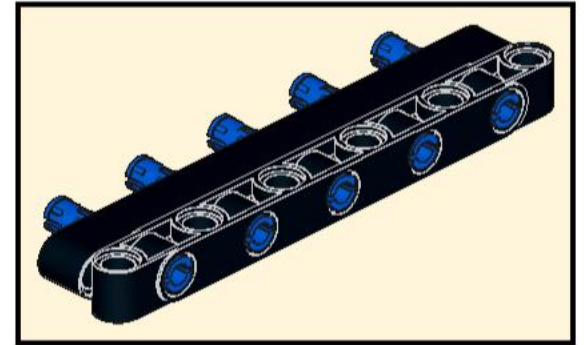
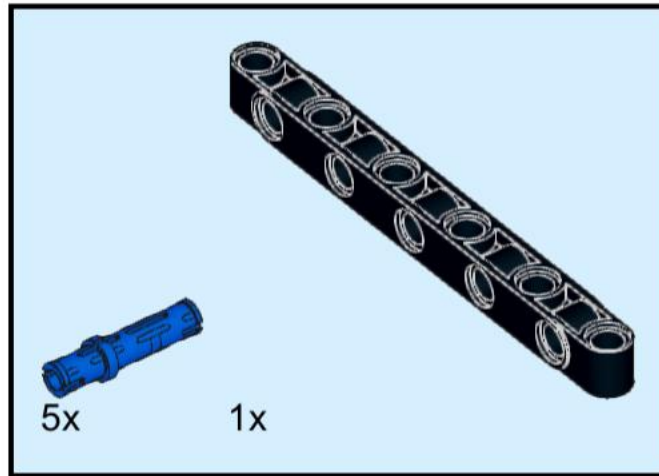
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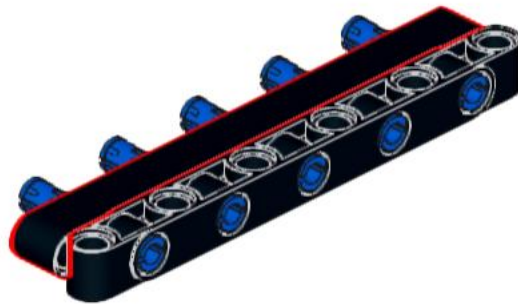
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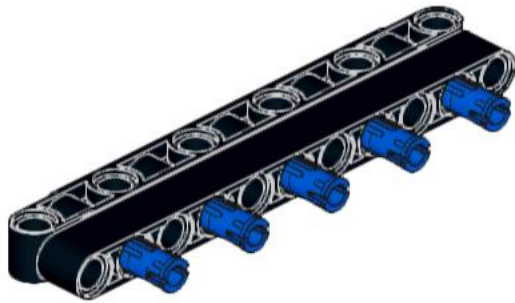
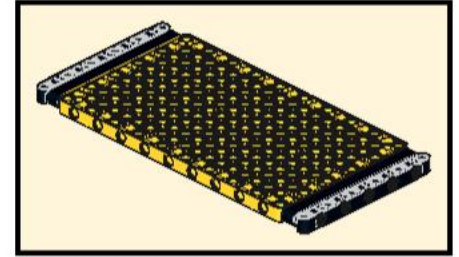
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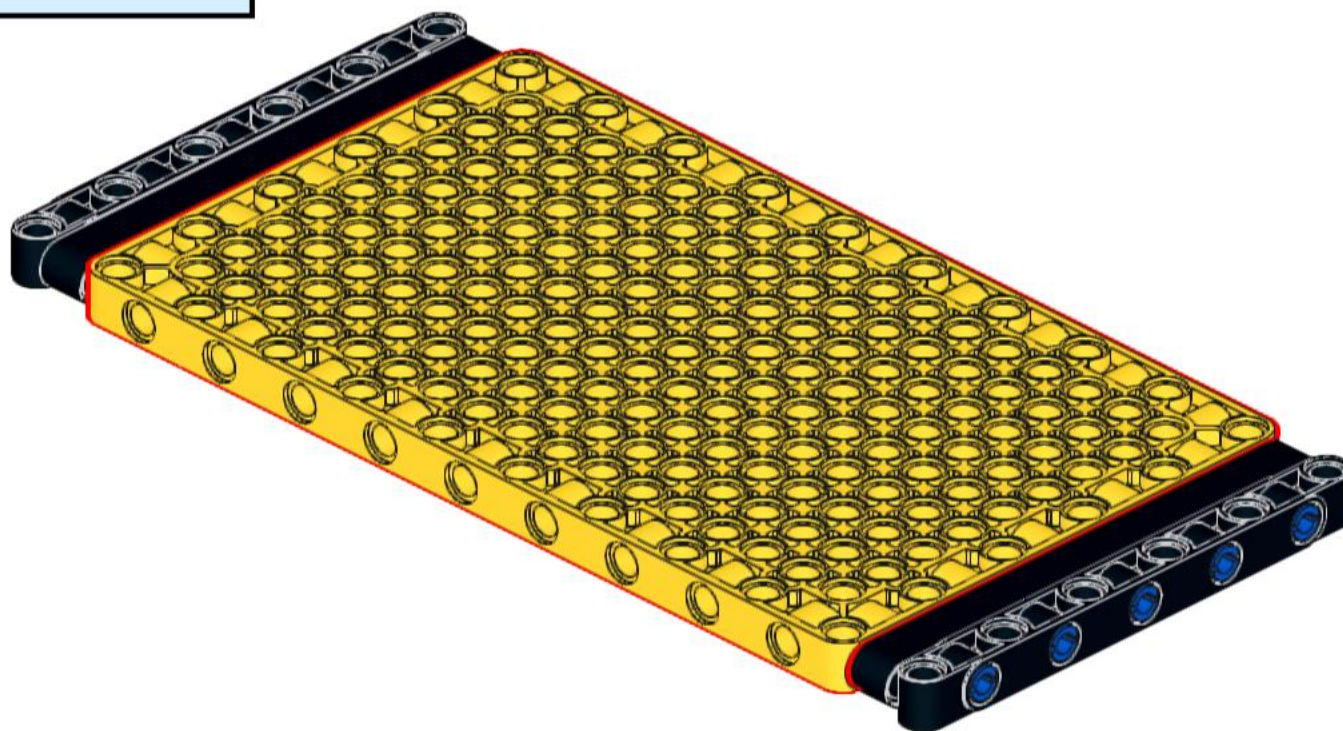
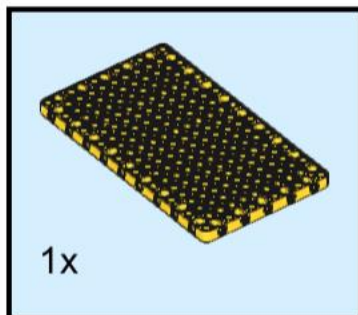
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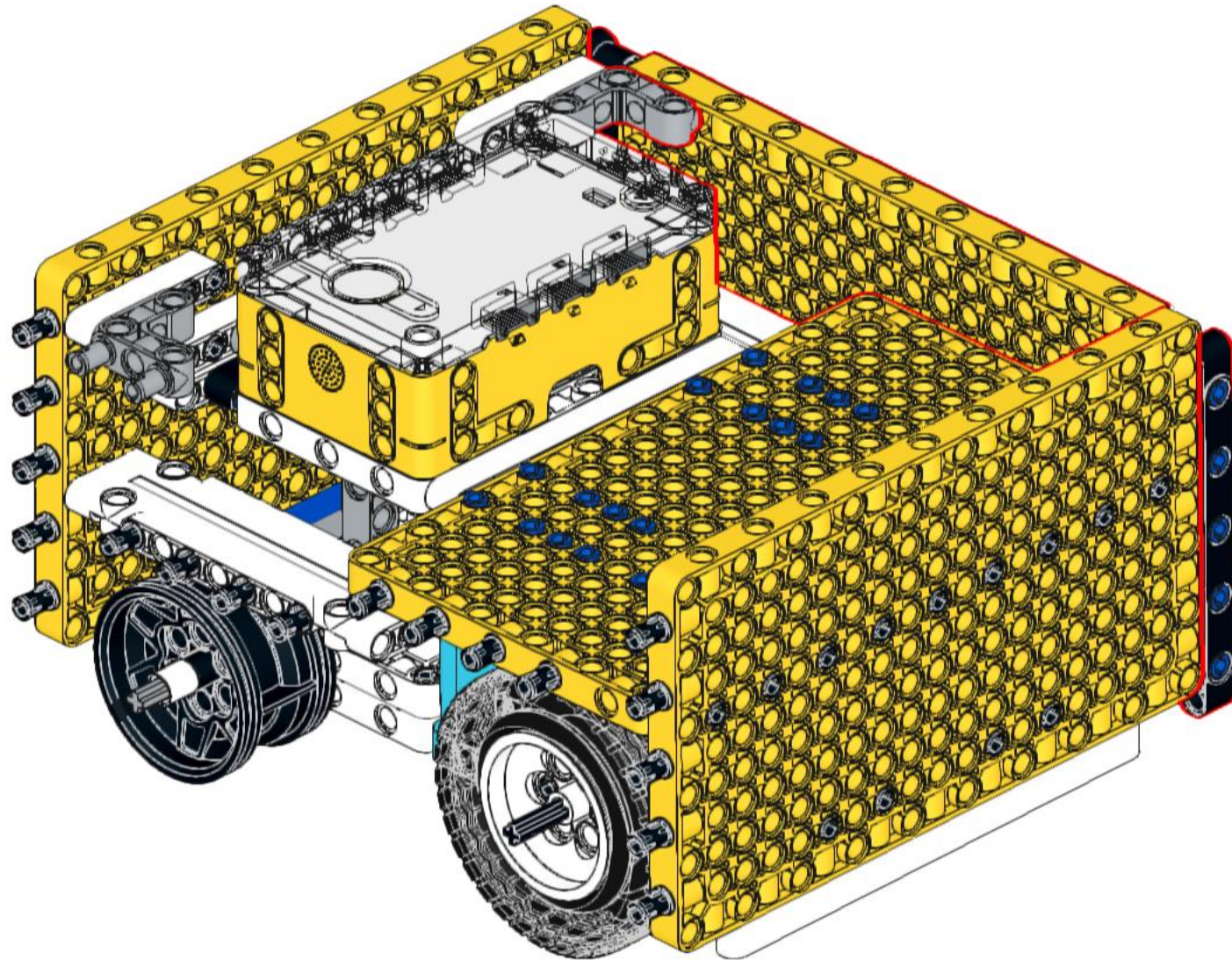
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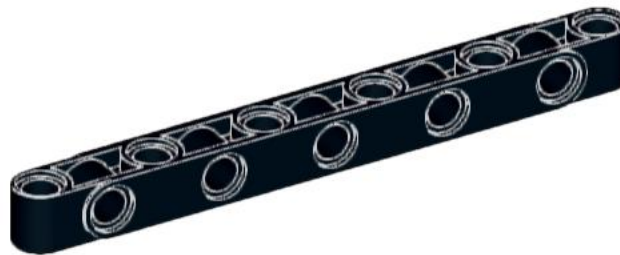
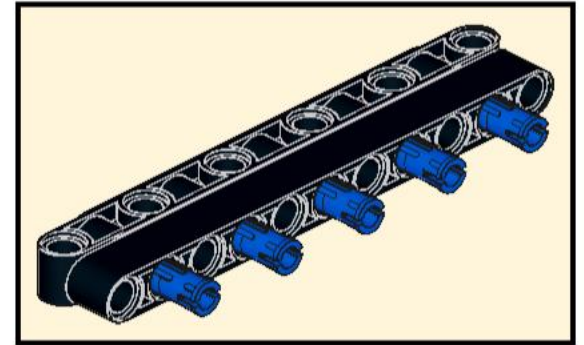
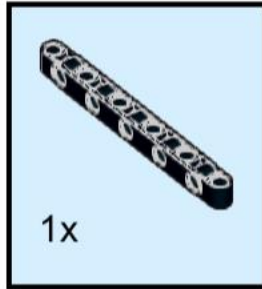
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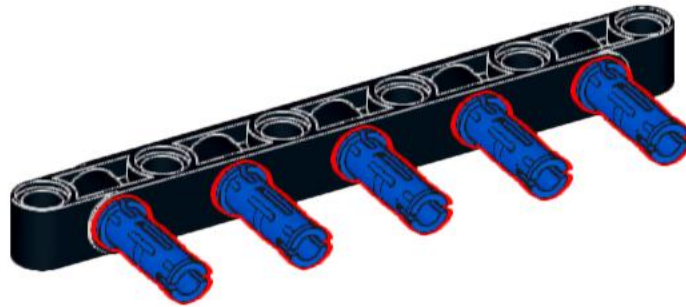
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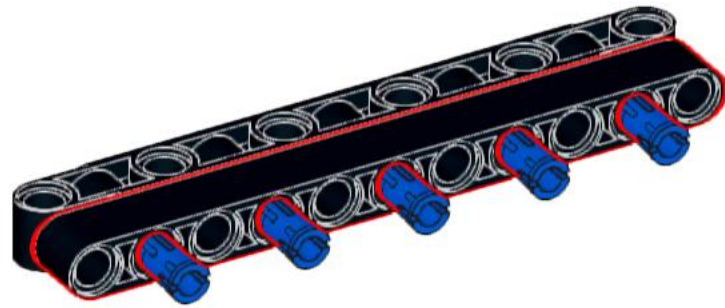
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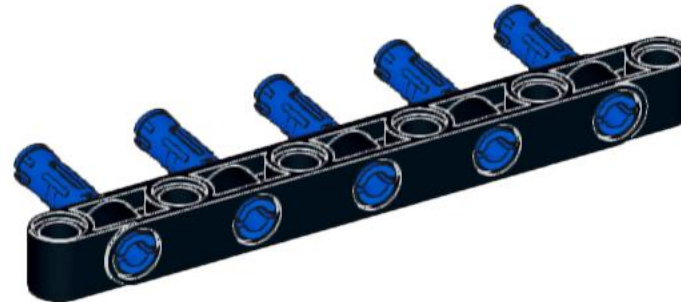
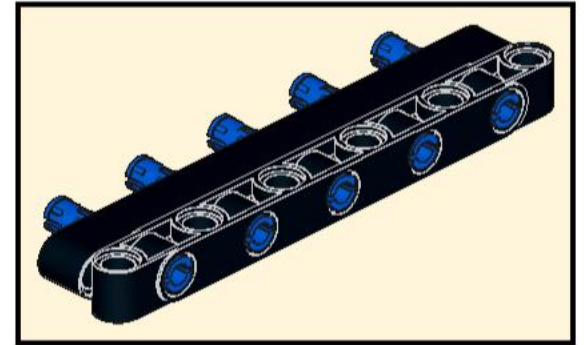
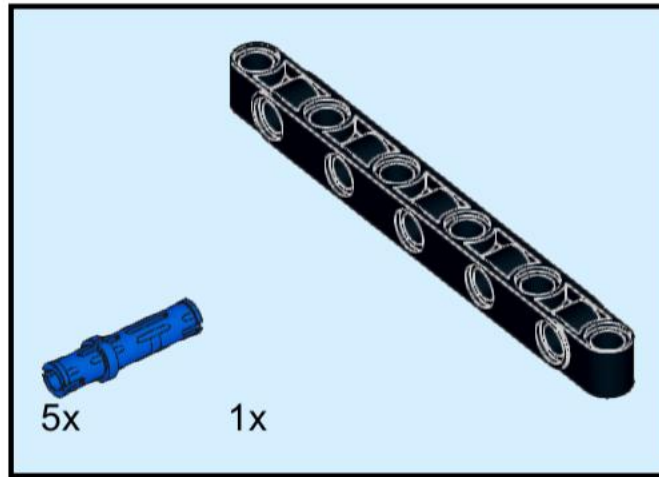
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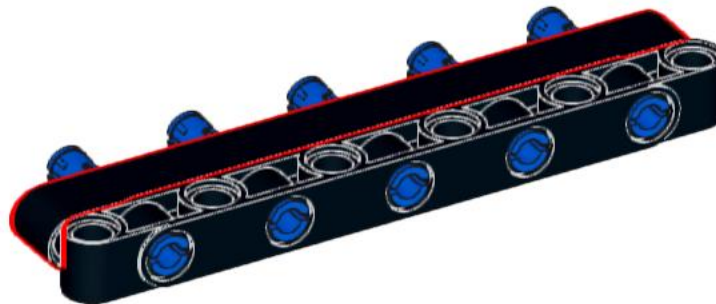
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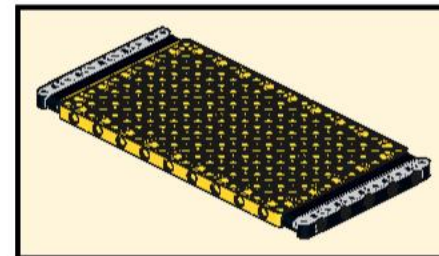
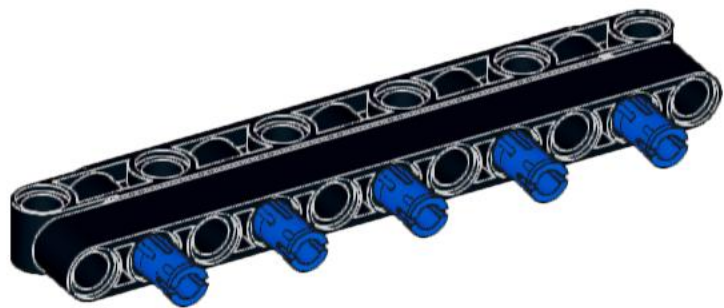
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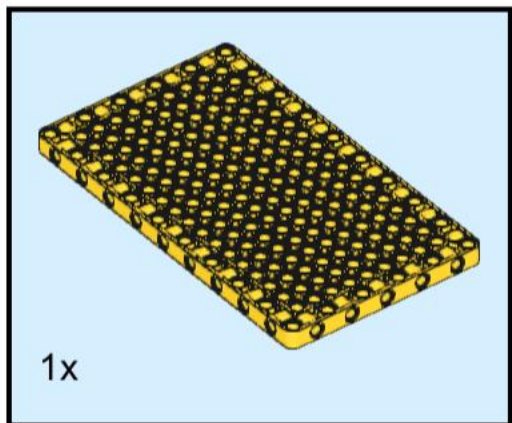
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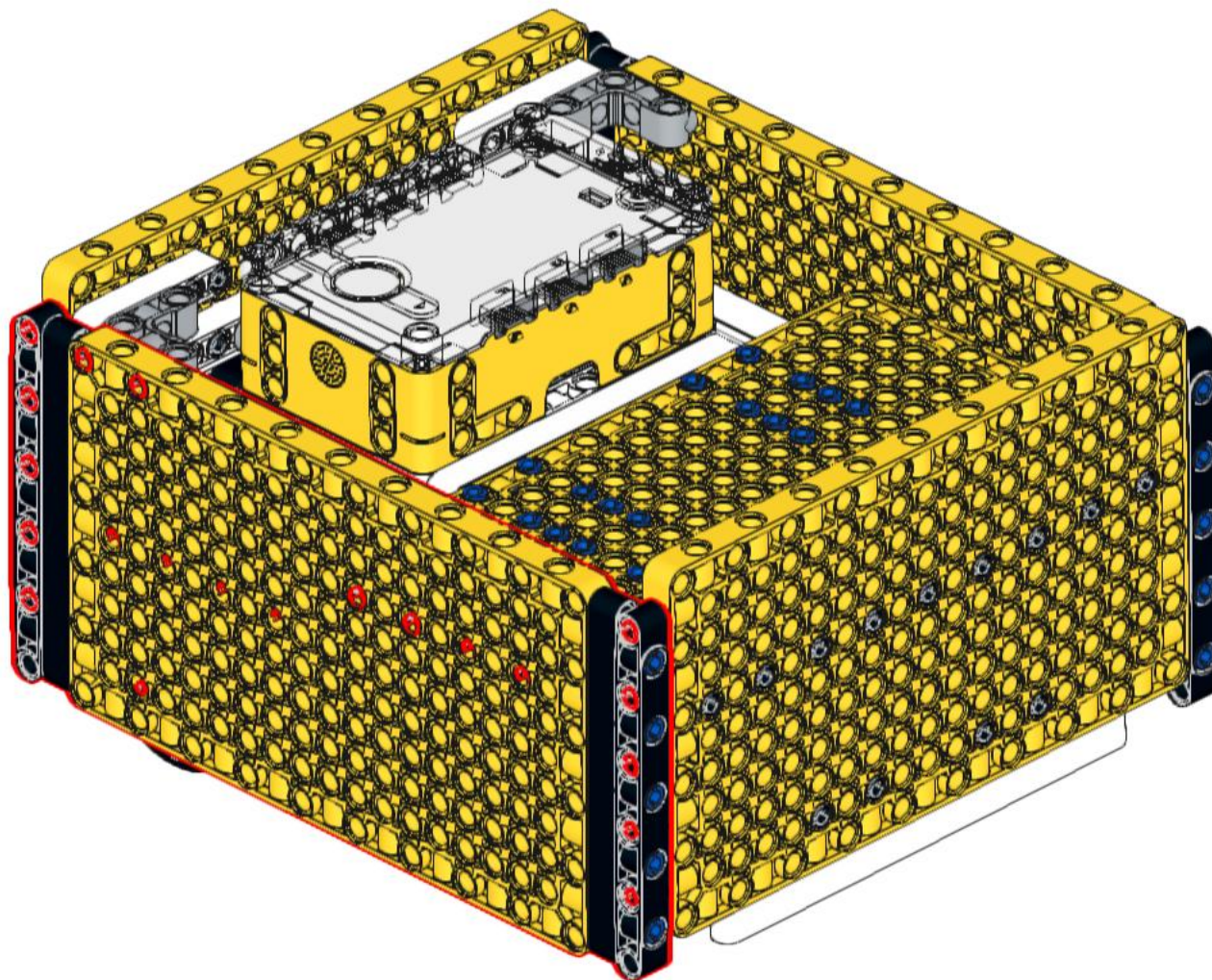
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The core box robot is completed.

The next step is to install your motors. We recommend using the large Spike motors due to substantially better torque over the medium Spike motors.

There are a substantial number of ways you can install the motors based on how the team wants its attachments to work.